



Chaotic dung beetle optimization–enhanced multi-agent deep reinforcement learning for joint task offloading and resource allocation in multi-unmanned aerial vehicle internet of medical things networks

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Abstract

The rapid growth of connected medical devices generates massive volumes of heterogeneous health data that must be processed and transmitted in real time. In such environments, minimizing latency and energy consumption remains a critical challenge for next-generation health monitoring systems. Existing reinforcement learning and optimization methods for intelligent communication networks face several challenges, including slow convergence, high computational overhead, and inefficiency in handling task prioritization. To resolve these issues, this work develops a chaotic dung beetle optimization-boosted multi-agent deep reinforcement learning that jointly optimizes communication reliability, computational efficiency, and task prioritization. A reward function is designed to jointly minimize delay, energy usage, and system cost while preserving information freshness. Specifically, the dung beetle optimization process is combined with a piecewise linear chaotic map to enhance population diversity, which significantly improves search space exploration and leads to faster convergence and higher solution quality. The proposed algorithm enhances the exploration capability of multi-agent deep reinforcement learning through the integration of chaotic dung beetle optimization, enabling more accurate and reliable decision-making in real-world applications. Extensive experiments demonstrate that the proposed chaotic dung beetle optimization-boosted multi-agent deep reinforcement learning model achieves superior performance compared to baseline algorithms. Specifically, it reaches an accuracy of over 97.00% with rapid convergence, reduces system cost under varying health data sizes and Medical Internet of Things devices, and maintains robust scalability across diverse workloads. Moreover, the model achieves significant reductions in communication latency and energy consumption as central processing unit cycles and bandwidth increase, while effectively prioritizing high-criticality tasks.

Keywords Chaotic dung beetle · Internet of Medical Things · Multi-agent deep reinforcement learning · Resource allocation · Unmanned aerial vehicle

1 Introduction

The Internet of Medical Things (IoMT) is widely applied to collect healthcare data of an individual human being to perform timely treatment. This plays a significant role in the healthcare field between devices and human beings. The delay of report generation in the healthcare field is lowered due to the advent of fifth generation (5G), which provides smart services and minimizes the delay (Jena et al. 2024). The IoMT transmits healthcare data effectively in real-time applications through wireless transmission. In the healthcare industry, effective end-to-end services are provided by accurately interconnecting the personal details of

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individuals and maximizing the speed of report generation. This provides a superior service by maximizing the flexibility and scalability with lower cost, as well as enhancing the quality of life. However, the significant issues obtained in IoMT resource allocation are privacy, delay, and network congestion (Tang et al. 2023). The resources are allocated by mobile edge computing (MEC), and the resource management improves the quality of service (QoS). In the 5G network, the unmanned aerial vehicle (UAV) is a significant technology in the 5G network that has gained more popular in various sectors. The modern healthcare industry relies on UAVs that securely collect and transmit data to the base station (BS). It also forces the IoMT device to function properly in various resources that control and monitor the formation of burden in data transmission (Rezaee et al. 2023). The resource allocation of IoMT in UAVs optimized the bandwidth and transmission power by meeting the requirements of the healthcare field. The usage of UAVs is to collect information and securely transmit it to the network by allocating the resources randomly (Dong et al. 2023).

The UAV assists the network by monitoring health data, and the resource allocation ensures the reliability of the healthcare scenario. The transmitted data are optimized and provide a quick response within the allotted time (Hasan et al. 2022). The delay is lowered by adjusting the resources in IoMT devices after computational servers are overloaded. In real-world applications, it is highly suitable and processed with limited resources, in which diagnoses are undertaken based on a smart healthcare system. The congestion in the network is also lowered by UAVs, and the edge servers are computed with heterogeneous services by collecting global data through the network (Lakhan et al. 2022). The IoMT devices distribute the task in smart healthcare applications and enhance the reliability of the wireless 5G network (Khan et al. 2022). The communication is performed in a standardized way by a UAV that lowers the congestion issues and transmits multiple data points to the network simultaneously (Osman 2024). The IoMT comprises diverse tools, namely sensors, wearable devices, and clinical systems that provide high-quality care to elderly people. It differs from other devices and reduces the formation of impacts by protecting the data with better privacy. It entails better security by providing integrity, authentication, and data confidentiality that constructs optimal dynamic treatment by solving real-world optimization issues (Consul et al. 2024). The IoMT lowers the latency, particularly in emergency cases, and attains a better volume of data for computational servers. The UAV-enabled IoMT network diminished the delay and processed the medical data within the allotted time by overcoming the posed challenges in the healthcare industry (Precup et al. 2025).

The rapid growth of UAV-assisted IoMT applications has created an urgent need for efficient and reliable healthcare data delivery. In such safety-critical environments, challenges such as highly dynamic network conditions, strict latency requirements, limited energy resources, and the need for accurate decision-making need to be addressed simultaneously. Existing methods, however, often optimize only a subset of these objectives, suffer from slow convergence, or rely on heuristic approaches that lack adaptability in real-world scenarios. To resolve these limitations, this work proposes the chaotic dung beetle optimization-boosted multi-agent deep reinforcement learning (ChDBO-MADRL), which integrates ChDBO with MADRL. This hybrid approach ensures robust adaptability to dynamic conditions and enables joint optimization of latency, energy, accuracy, and system cost, making it highly suitable for practical IoMT healthcare applications where timely and reliable decisions directly impact patient safety. While many existing metaheuristics could in principle be adapted for parameter tuning, the motivation for introducing ChDBO lies in its natural ability to balance exploration and exploitation through biologically inspired search patterns. By embedding chaos into the dung beetle optimization algorithm (DBO) framework, population diversity is increased, helping to avoid premature convergence that often limits traditional algorithms such as particle swarm optimization (PSO), genetic algorithm (GA), or grey wolf algorithm (GWA) in dynamic and multi-objective environments. Furthermore, the full structure of ChDBO, rather than just a single operator, is required to reliably guide RL training toward robust link and resource allocation policies. Therefore, ChDBO is not just a minor operator adjustment but a targeted optimization scheme designed for the specific challenges of UAV-assisted IoMT.

1.1 Novelty

The ChDBO-MADRL is a innovative novel approach that solves the challenges encountered in healthcare services based on 5G network. The novelty of the work is presented as follows.

- *Chaos-enhanced exploration* This work introduces ChDBO to enhance exploration diversity, accelerate convergence, and avoid local optima in multi-agent learning.
- *Joint resource-task optimization* This work designs a unified MADRL framework that simultaneously optimizes task offloading, central processing unit (CPU) allocation, transmit power, and bandwidth for UAV-enabled IoMT networks.
- *Hybrid training strategy* The proposed model employs centralized critic training with decentralized actor

execution to achieve both scalability and coordinated decision-making among UAVs.

- *Healthcare applicability* This work demonstrates practical use in UAV-enabled healthcare systems, where MEC and UAVs collaboratively process and transmit patient data with improved scalability, flexibility, and low latency.

1.2 Contribution

The significant contributions of the article are deliberated as follows.

- *Joint optimization framework* Most prior works address either task offloading or resource allocation independently, often leading to sub-optimal end-to-end performance. This work combines task offloading, CPU allocation, transmit power control, and bandwidth assignment into a single learning framework, enabling coordinated decision-making and improved global optimization.
- *Chaos-enhanced metaheuristic for exploration* Traditional frameworks often encounter slow convergence and local optima due to limited exploration capability. ChDBO with Piecewise Linear Chaotic Map (PWLCM) is employed to improve diversity in search space exploration, outperforming other metaheuristic-enhanced RL methods in convergence speed and solution quality.
- *Centralized training with decentralized execution* Prior DRL-based resource allocation works adopt fully centralized or fully decentralized learning, which either limits scalability or sacrifices coordination. Centralized critic training with decentralized actor execution is employed, ensuring scalability, reduced communication overhead, and effective coordination among the UAV cluster head (UCH).
- *Superior performance on multiple metrics* Across all existing algorithms, the proposed ChDBO-MADRL model consistently achieves superior performance, particularly in terms of accuracy.

The remaining sections of the paper are as follows: In Sect. 2, the related works based on resource allocation and optimization are discussed. The system model and problem formulation are emphasized in Sect. 3, and in Sect. 4, the proposed work is explained. The results, graphical analysis are presented in Sect. 5. The conclusion of the article and future works are provided in Sect. 6.

2 Related works

This section provides a summary of various literature works based on resource allocation and optimization in UAVs, as well as trajectory optimization in UAV-enabled IoMT networking systems.

2.1 Resource allocation and optimization in UAV-enabled IoMT networks

Seid et al. (2023) implemented the multiagent federated RL (MAFRL) for resource allocation in UAV-enabled IoMT. A distributed algorithm was integrated into the model to offer distributed computing. An extensive analysis exhibited that the MAFRL model attained better performance. Gao et al. (2024) developed a UAV-enabled distributed computation offloading and power control approach with multi-agent (DOPMA). The offloading and power control decisions were derived by implementing an improved multi-agent deep deterministic policy gradient algorithm. The experimental results proved that the performance of a developed model was superior.

For effective resource allocation, Nahom et al. (2023) presented the federated deep RL (FDRL) method. The IoMT device was related to aerial flying BS (AFBS) based on the data quality, where the accuracy and learning efficiency were improved. Finally, the broad analysis demonstrated that the developed FDRL model attained higher accuracy and lower computational cost. Tran et al. (2021) implemented the UAV relay-assisted emergency communications in IoT networks for resource allocation and trajectory optimization. The major objective was to integrate the UAV trajectory while satisfying the limited storage capacity of the UAV. Thus, the simulation results revealed that the developed model achieved higher throughput and the number of served IoT devices.

For data streams, Jaworski et al. (2012) explored a resource consumption of the fuzzy clustering algorithm. Here, two algorithms were analyzed to evaluate their performance in resource optimization, such as the fuzzy C-means algorithm (wFCM) and the fuzzy possibilistic C-means algorithm (wPCM). This work primarily focused on improving clustering efficiency in the presence of evolving data distributions. For feature selection, Kilic et al. (2023) presented an Anarchic Society Optimization algorithm. This work demonstrated the effective performance of the algorithm against other optimization algorithms. Performance assessments were conducted on a set of benchmark datasets to ensure the performance of the algorithm.

For the classification of stationary data streams, Jaworski et al. (2017) developed a family of new splitting criteria. Here, refined splitting rules were formulated using

Gini index impurity measures and misclassification error. In this work, the type-1 splitting criteria served as a key factor for the selection of attributes. Experimental results demonstrated improved robustness of the model. For constructing local volatility surfaces, Zheng-Liang and LOK (2024) developed a dimension-reduced modeling approach. This work focused on addressing the high complexity and computational cost of volatility surface modeling. The work demonstrated that unsupervised learning effectively improves both model efficiency and accuracy.

2.2 Security and trajectory optimization in UAV-enabled IoMT system

Asim et al. (2023) developed a simulated annealing trajectory optimization algorithm (SATO) in an autonomous UAV-powered mobile fog computing (MFC) system for medical Internet of Things (IoT) devices. In this paper, SATOA was implemented to design the UAVs' trajectories. The stop points were updated by utilizing different population sizes. The experimental results revealed that the developed SATOA was the better solution. Zhou et al. (2024) presented the joint uplink and downlink rate splitting for fog-computing-enabled IoMT. The optimal computing resource allocation was achieved by using closed-form expressions. Lastly, the intrinsic and auxiliary variables were updated by applying the alternating optimization algorithm. Thus, an extensive analysis exhibited that the transmit scheme gained better performance. Ghazal (2021) presented the positioning of UAV BSs. In this paper, the Unified Greedy Quest Algorithm for the IoMT (UGQA-IoMT) was utilized for optimal places. The UGQA-IoMT was used in different conditions, where the UAVs were installed by themselves or with the set BS. An experimental analysis proved that the developed UGQA-IoMT attained a dispersed overall load balance nearer to 1.

3 Research gap and limitations

Existing methods show promising results under specific conditions but are unsuccessful when applied to large-scale, dynamic, and delay-sensitive healthcare environments. These challenges highlight the need for more robust and adaptive solutions. The main limitations of existing methods and how our proposed ChDBO-MADRL framework addresses them are summarized below.

- *Limited adaptability* Conventional optimization algorithms rely on static parameter tuning and fail to adapt to the dynamic wireless environment of IoMT systems. The proposed system enables ChDBO to maintain

population diversity, avoid premature convergence, and accelerate training.

- *Premature convergence* Many metaheuristic and DRL-based models converge quickly to sub-optimal solutions, minimizing system efficiency in large-scale deployments. The proposed model performs *joint optimization* of task offloading, bandwidth allocation, transmit power, and CPU cycles, demonstrating balanced system performance across multiple objectives.
- *Separated optimization of tasks and resources* Existing works frequently optimize either task offloading or resource allocation independently, resulting in unbalanced system performance and inefficient resource utilization. The proposed model utilizes a centralized training with a decentralized execution strategy, which enables stable learning while maintaining scalability in multi-UAV networks.
- *Scalability issues* Centralized learning-based methods struggle to coordinate a growing number of UAVs and IoMT devices without incurring excessive overhead. The proposed model leverages the actor-critic structure of MADRL, guided by ChDBO, to speed up convergence and deliver near-optimal policies in fewer episodes.
- *Slow convergence of RL models* Standard RL approaches require a large number of episodes to stabilize, making them less practical for real-time healthcare scenarios. Ensuring practical relevance in healthcare IoMT, where the framework supports low-latency, energy-efficient, and scalable medical data transmission and processing.

4 System model and problem formulation

The multi-agent air-to-ground (ATG) network is employed with energy models, communication, as well as computation. In the field of healthcare systems, the multi-agent ATG network is determined based on computation offloading, association, and reliable resource allocation, which is represented in Fig. 1. In this, the health condition of the patient is frequently monitored by smart devices, namely smart watches, sensors, etc., which is obtained in the lower layer of the IoMT device. Moreover, the edge computing nodes such as BS, intelligent ambulance, as well as UCH are used to extend the resource network, and stability of the network is maintained effectively by Software Defined Networking (SDN) that handles resource management, computation offloading, network infrastructure, and association.

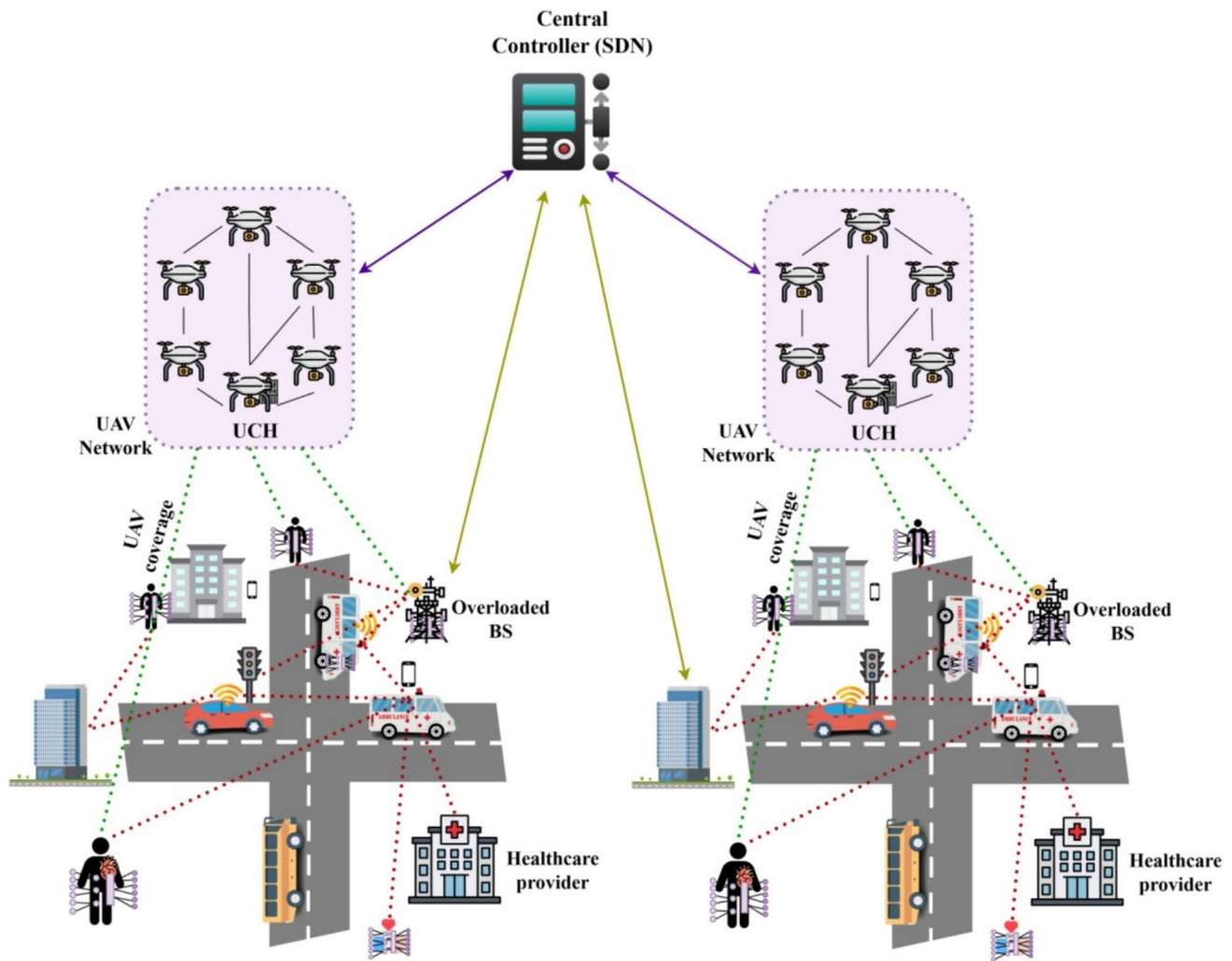


Fig. 1 UAV-assisted healthcare framework

4.1 Communication model

The UAV-enabled ground network is interconnected with various smart devices that are implemented with MEC servers controlled by UCH, which generate multiple Medical IoT Devices (MID). This is used to validate patients' health conditions frequently and generates a report updated from the healthcare system. It randomly distributes Q small cells equipped with BS H and T but a few cells are obtained far away from the network. The formulation for BS and MID is indicated by $H \triangleq \{h = 1, 2, \dots, H\}$, $T \triangleq \{t = 1, 2, \dots, T\}$, where h is the index representing a specific BS and t is the index representing a specific BS. K clusters determine the clustering of the UAV network. Each and every cluster is equipped with UAVs at an altitude $S_U > 0$, where S_U specifies the altitude of the UAV. The sets of clusters and UAVs are expressed as $K \triangleq \{k = 1, 2, \dots, K\}$,

$U_k \triangleq \{u_k = 1, 2, \dots, U\}$, where k denotes the index representing a specific UAV cluster, U represents the total number of UAVs in each cluster, U_k denotes the set of UAVs in cluster K , and u is the index representing a specific UAV within cluster K . This article determines the ground network gathering support from UAVs, and this occurs when the BS malfunctions while resulting in healthcare issues. In robustness analysis, the ground network failed to reach urban areas, so it is necessary to extend the network coverage for better communication. The offloading of MID data is performed via either BSs or UAVs, which transmit the data through wireless links to medical servers for processing. To support this, the system operates in discrete time slots, indexed by $p \in P$, where $P \triangleq \{p | p = 1, 2, \dots, P\}$, and P is the total number of time slots.

To minimize transmission loss, the network is modelled in a 3D Cartesian coordinate system, where each BS position is denoted by coordinates (x_h, y_h, z_h) , each UAV in the

cluster K is indicated by (x_U, y_U, z_U) , and each MID is denoted by (x_j, y_j, z_j) . The following equation represents the Euclidean distance between a UAV and MID:

$$d_{U,t} = \sqrt{(x_U - x_t)^2 + (y_U - y_t)^2 + (z_U - z_t)^2} \quad (1)$$

In the above equation, $d_{U,t}$ is the Euclidean distance between UAV U and MID t , x_U, y_U , and z_U are the Cartesian coordinates of UAV U . x_t, y_t , and z_t are the Cartesian coordinates of MID t .

4.1.1 Medical emergent data

In healthcare services, multiple types of health records are generated based on high and low severity levels. The health data collected through MID are distinguished with varied significant classes. After extracting MID data, the medical tasks are updated and allocated resources regarding priority level. Each MID task is modelled as $(D_t, T_t^{\max}, G_t)$, where D_t indicates the input data size, $T_t^{\max}$ specifies the maximum tolerable latency, G_t is the required CPU cycles per bit. These tasks represent medical emergent data, where delay or packet loss results in severe degradation of patient care. Hence, efficient offloading and processing strategies are critical.

4.1.2 Age of information

The monitoring of patients' data accurately is a challenging task, and the freshness of the information is analyzed by the Age of Information (AoI). In offloading condition, the MID frequently serves the medical server node, which helps to update the medical records to MEC accurately and allotted time. The resources are effectively allocated, and the patient data generation process enhances the effectiveness with lower latency.

$$\Delta_t(P) = P - U_t \quad (2)$$

In the above equation, $\Delta_t(P)$ indicates the AoI of the device t at slot P and U_t is the most recent time slot in which a successful update from t was received. A lower AoI indicates fresher medical information, which is important for the timely detection of critical health events. Thus, the proposed framework reduces AoI alongside energy and delay.

4.1.3 Local computation

The MID analyzes the offloading task with the resources that compute the intensive task to the edge server by allocating resources to the local task. Then the edge servers process the

offloaded task and provide the outcome by computing the medical data, and the completion time of the local task. If a task is executed locally by MID t , the execution time and energy consumption are:

$$T_t^{\text{local}} = \frac{D_t \cdot C_t}{f_t} \quad (3)$$

$$E_t^{\text{local}} = K f_t^2 D_t \cdot C_t \quad (4)$$

In the above equations, f_t represents the CPU of the MID t , K indicates the effective CPU capacitance coefficient, T_t^{local} specifies the local execution time, and E_t^{local} indicates the local execution energy. Local processing is limited by low device capability and battery constraints, which motivates computation offloading to UAVs and MEC servers.

4.1.4 MID medical data offloading model

The offload task of UCH and BS gathers the information from MID v by validating the time delay cost by three factors, such as execution time, transmitting time as well and outcome delay. When local resources are insufficient, MID offloads tasks to UAVs or MEC servers via wireless links. The achievable transmission rate from the MID t to the UAV U is given in the following Eq. (5). For data transmission, each MID is assigned a bandwidth B_t , subject to noise power N_0 , and inference T_t .

$$R_{t,U} = B_t \log_2 \left(1 + \frac{P_t h_{t,U}}{N_0 + T_t} \right) \quad (5)$$

In the above equation, $R_{t,U}$ is the achievable transmission rate, $h_{t,U}$ indicates the channel gain between MID t and UAV U , P_t indicates the transmit power of MID t . The total offloading delay is indicated in the following equation:

$$T_t^{\text{off}} = \frac{D_t}{R_{t,U}} + \frac{D_t \cdot C_t}{f^{\text{MEC}}} \quad (6)$$

In the above Eq. (6), f^{MEC} is the computation capacity allocated by the MEC server, and T_t^{off} is the total offloading delay. By combining UAV-assisted relaying with MEC computing, the system demonstrates low latency, reduced energy consumption, and improved scalability, which are essential for large-scale IoMT healthcare deployments.

4.2 Energy consumption

Energy consumption in the UAV-enabled IoMT network arises from four primary sources such as task transmission, local execution, UAV hovering, and UAV flying. These

components jointly determine the total energy required for task offloading and processing. During each time slot, the transmission energy for MID t is expressed as:

$$E_t^{tx} = P_t \cdot \frac{G_t}{R_{t,U}} \quad (7)$$

In the above equation, E_t^{tx} is the transmission energy consumed by the MID t , G_t is the size of the task data to be offloaded by the MID t . This term captures the energy required to send medical data to UAVs and MEC servers. If a task is processed locally, the energy depends on the MID's CPU characteristics. This indicates the quadratic relation between CPU frequency and energy consumption. When a task is offloaded, the UAV executes the task using its computation capacity. The computation energy E_t^{MEC} at the UAV follows:

$$E_t^{MEC} = K_{MEC} (f_t^{MEC})^2 D_t C_t \quad (8)$$

In the above equation, K_{MEC} is the CPU coefficient of the MEC server, and f_t^{MEC} is the allocated CPU frequency. The propulsion energy of UAVs comprises hovering and flight components:

$$E_U^h = P_U^h \cdot T_U^h \quad (9)$$

$$E_U^F = P_U^F (v_U) \cdot \frac{S_U}{v_U} \quad (10)$$

In the above equations, P_U^h denotes the hovering power of the UAV U , T_U^h denotes the hovering time, S_U indicates the flight distance of the UAV U , v_U is the flight speed, E_U^F is the flight energy consumed by the UAV U , P_U^F is the flight power consumption of the UAV U . For each task, the total energy requirement is given in the following equation:

$$E_t^{tot} = E_t^{tx} + E_t^{local} + E_t^{MEC} + E_U^h + E_U^F \quad (11)$$

In the above equation, E_t^{tot} is the total energy consumed in the processing task tk , considering communication, computation, and UAV propulsion. This unified model highlights that energy efficiency cannot be achieved by optimizing only one component. Therefore, the proposed framework jointly considers communication, computation, and mobility energy to strike a balance between task delay, UAV endurance, and healthcare data freshness.

4.3 Problem formulation

In UAV-assisted IoMT networks, medical tasks generated by heterogeneous devices must be executed under

constraints on latency, energy efficiency, and data freshness. Direct execution on IoMT devices is limited by scarce computational resources, while offloading to UAVs or MEC servers introduces additional communication delays and energy consumption. Additionally, the highly dynamic wireless environment and mobility of UAVs make resource allocation and task scheduling even more challenging. To systematically address these issues, it is important to cast the task execution and resource management process as an optimization problem that jointly considers latency, energy, and information freshness, while respecting the system constraints on bandwidth, CPU capacity, and transmission power. For each task i , a binary variable y_i determines whether it is processed locally ($y_i = 0$) or offloaded ($y_i = 1$). If the task i is offloaded, the total delay includes transmission and computation times:

$$T_i = \frac{T_i}{r_i} + \frac{C_i}{f_i} \quad (12)$$

In the above equation, T_i is the task time, r_i is the transmission rate, f_i is the allocated CPU frequency, and the total number of CPUs is denoted as C_i . For local execution, latency is provided in the following equation:

$$T_i = \frac{C_i}{f_i} \quad (13)$$

The energy consumption includes transmission and computation: The goal is to minimize the weighted system cost that combines latency, energy, and AoI. The objective function is given in Eq. (15)

$$E_i = p_t \cdot \frac{T_i}{r_i} + K C_i f_i^2 \quad (14)$$

$$\min J = \alpha \sum_i T_i + \beta \sum_i E_i + \gamma \sum_i \Delta_j \quad (15)$$

In Eq. (15), the energy consumed for the *computing* task is represented as $K C_i f_i^2$, J is the overall system cost, Δ_j is the AoI for the task i , and α , β , and γ are design weights reflecting system priorities.

4.3.1 Constraints

The formulated optimization is subject to several practical constraints. First, the task deadline constraint ensures that the latency of each task does not exceed its maximum allowable delay δ_i , which is critical in healthcare applications. Second, the CPU capacity constraint enforces the limited computational capability of IoMT devices, UAVs, or

MEC servers, thereby preventing resource over-allocation. Third, the bandwidth limitation constraint captures the scarcity of wireless spectrum resources, ensuring that the total allocated bandwidth does not exceed the available capacity. Finally, the power budget constraint restricts the transmission power of IoMT devices within hardware and regulatory limits, which is essential for prolonging device lifetime and maintaining safe operation in the network.

- Task deadline:

$$T_i \leq \delta_i, \forall i \quad (16)$$

- CPU capacity:

$$\sum_i f_i \leq F^{\max} \quad (17)$$

- Bandwidth limitation:

$$\sum_i b_i \leq B^{\max} \quad (18)$$

- Power budget:

$$0 \leq p_i \leq P^{\max} \quad (19)$$

In the above equations, $F^{\max}$ is the maximum available CPU capacity, b_i is the bandwidth allocated to the task i , $B^{\max}$ is the maximum total available system bandwidth, p_i is the transmission power, and $P^{\max}$ is the maximum transmission power limit of an IoMT device. Rather than applying a traditional penalty function-based mathematical solver, the constraints are integrated into the reward design and action space definition of the MADRL model:

- *Feasible action space* Actions (offloading decisions, CPU allocation, bandwidth, and transmit power levels) are generated so that they inherently satisfy basic bound constraints.
- *Reward shaping* The reward function is constructed to penalize violations of resource limits, QoS thresholds, and fairness constraints, thereby steering agents toward feasible solutions.
- *ChDBO search guidance* The ChDBO component adjusts exploration while demonstrating that candidate solutions comply with the problem constraints before being evaluated.

In the proposed ChDBO-MADRL framework, these constraints are integrated into the optimization process. Hard constraints (task deadlines, bandwidth, and CPU capacity)

are enforced by restricting the feasible action space, ensuring that infeasible solutions are not explored. Soft constraints such as energy consumption are incorporated into the reward function as penalty terms. This enables the agents to learn energy-efficient and balanced policies. Although the formulation in this work primarily considers box-type constraints, the proposed ChDBO-MADRL framework can also handle more general non-box constraints. In such cases, infeasible actions can either be projected back into the feasible region or penalized via the reward function. Moreover, the ChDBO component guides exploration to feasible subspaces by rejecting invalid candidate solutions. This ensures that the learning framework remains applicable even under coupled, nonlinear, or non-convex constraints commonly observed in real-world IoMT healthcare networks. It demonstrates that the post-processing or penalty projection is not needed to ensure constraint satisfaction, highlighting that it is carried out when training the model. This approach avoids producing infeasible solutions during both exploration and exploitation, while still enabling the algorithm to explore a wide feasible search space. The formulated joint task offloading and resource allocation problem in multi-UAV IoMT networks is expressed as a mixed-integer nonlinear programming model, which is Nondeterministic Polynomial-time hard (NP-hard) and involves complex coupling between discrete and continuous decision variables. MADRL is employed to overcome the sequential and distributed nature of the decision-making process, enabling UCH to coordinate resource allocation and offloading strategies in dynamic environments. However, standard RL often suffers from slow convergence and sub-optimal solutions due to limited exploration. To overcome this, ChDBO-MADRL is combined to enhance the exploration of the solution space and guide the learning agents toward high-quality solutions. This hybrid design demonstrates that the algorithm directly targets the minimization of latency and energy consumption specified in the optimization problem while satisfying the bandwidth, power, CPU, and fairness constraints.

5 Proposed methodology

The outline of the proposed model is depicted in Fig. 2. In this system, agents (Agent 1 to Agent n) receive observations from the environment, take actions, and obtain rewards based on their performance, while also performing agent-agent observation exchange to support cooperative decision-making. Each agent trains a local model on its respective healthcare or traffic-related data. These local models are then combined through Global Federated Learning (FL), where parameters such as $Q_1, Q_2, \dots, Q_n$ are exchanged and aggregated into a global model, ensuring

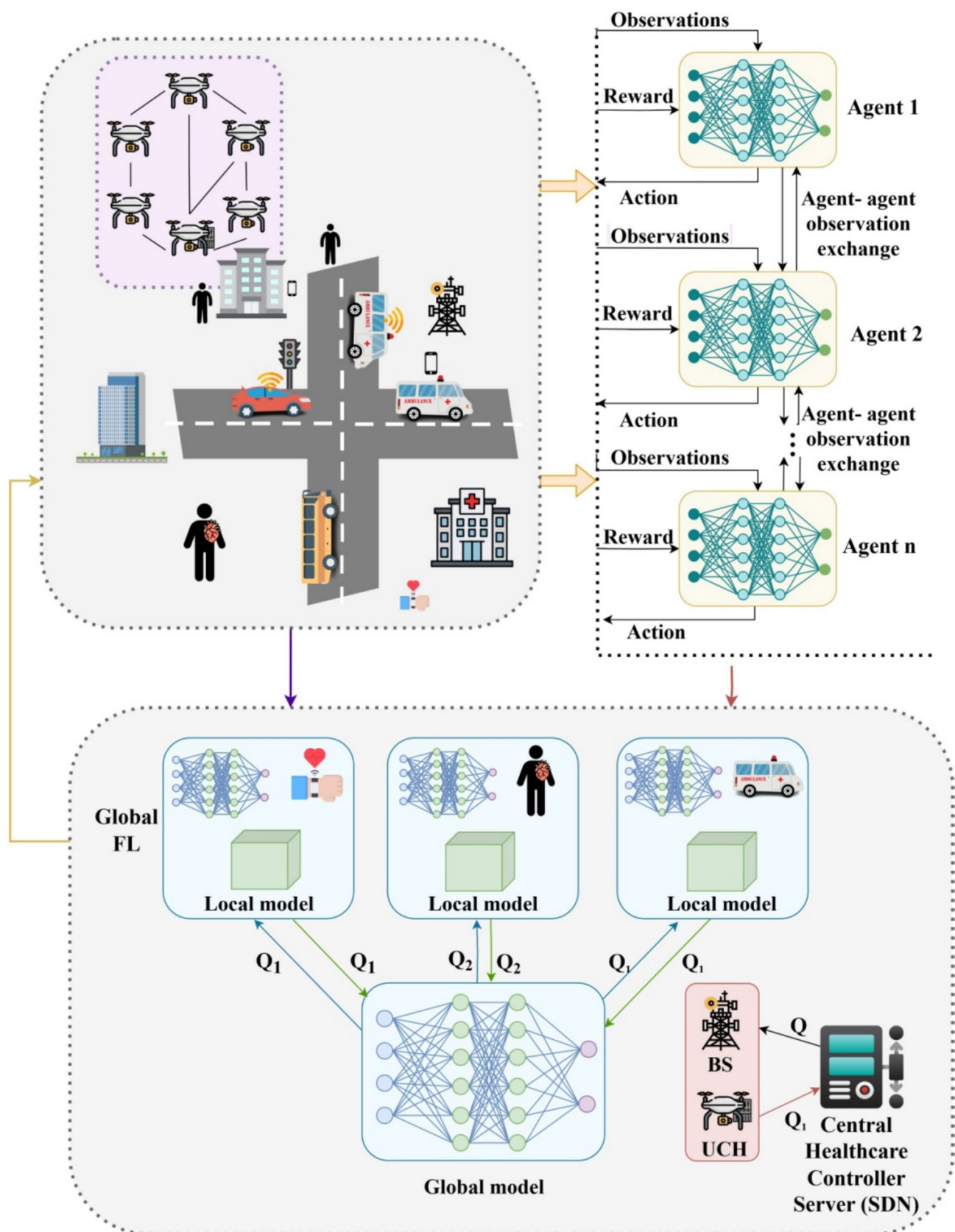


Fig. 2 Conceptual framework of the proposed model

data privacy by avoiding direct sharing of raw information. The SDN coordinates this process by receiving the local model updates from agents and connected infrastructure like BS and unmanned communication hubs (UCH), and then distributing the improved global model back to the agents. Through this coordinated exchange of observations, rewards, actions, and Q-values, the framework enables efficient ambulance routing, patient monitoring, traffic control, and drone-assisted communication, while maintaining security and enhancing overall healthcare intelligence.

5.1 Detection and classification in IoMT workloads

The IoMT devices continuously sense biomedical signals such as the Electrocardiogram (ECG). The detection task identifies the presence of clinically relevant events, while the classification task assigns the detected signal to the correct heartbeat class. These tasks are computationally intensive and cannot always be handled locally by low-power IoMT sensors. This is where the UAV-assisted MEC network plays a crucial role: the decision of whether to process locally, at the UAV, or at the edge server directly determines the timeliness and accuracy of detection/classification outcomes. The optimization problem formulated in the proposed framework, therefore, links these two perspectives:

- *Classification accuracy* depends on having enough computational resources and low transmission errors.
- *Detection latency* depends on communication link quality and offloading decisions.
- The reward function in ChDBO-MADRL integrates these factors: actions that minimize delay and energy while preserving or improving classification accuracy receive a higher reward.

Hence, the detection/classification problems are not side tasks but are embedded into the optimization objective itself: resource allocation and link decisions are optimized precisely to maximize the reliability and timeliness of these healthcare Artificial Intelligence (AI) tasks.

5.2 Multi-agent deep reinforcement learning

This research employs the MADRL system that optimizes the resource allocation and manages the decentralized decision-making in a dynamic and complex environment in a UAV-enabled IoMT healthcare network. The MADRL permits real-time decision making in various networking scenarios, thereby ensuring optimal performance. The detailed description of the MADRL algorithm is elaborated in the following subsection.

5.2.1 A. Modeling of multi-agent environments

The allocation of the radio resources and power levels is improved using RL by agents, which refines performance at the same time reduces interference with cellular users (Wang et al. 2024). In the proposed framework, the state space is denoted as Z , the action space is denoted as Y , and the reward function is denoted as u^a .

State space (Z): All agents monitor the following during the specified time step;

- *Own channel quality ($H_l^{b,a}$):* Evaluates how fine its transmitter as well as receiver can interact.
- *Cellular channel quality ($H_l^{e,a}$):* Evaluates interference due to the agents on the cellular network.
- *Past interference (L_l^{a-1}):* Monitors the interference amount experienced by the agents in earlier occurrences.
- *Previous radio resources (J_l^{a-1}):* Recall the radio resources utilized in the last time step.

These considerations from the state of the agent are mathematically represented as follows,

$$Z_l^a = [H_l^{b,a}, H_l^{e,a}, L_l^{a-1}, J_l^{a-1}] \quad (20)$$

In the above equation, Z_l^a represents the state vector of the agent l .

Action space (Y) During every time step, the agent selects an action ($y_l^a \in Y$) based on its current state ($z_l^a \in Z$). Choosing a radio resource to send on and modifying its transmission power are the actions included in this action space. This action space scales with the count of radio resources as well as energy levels available, which makes it adaptable for diverse network setups. Every agent's action selection must fulfill the constraint ($\tau_{n,j}^e \geq \tau_{MIN}^e$), where the Signal-to-Noise-Plus-Interference Ratio (SNIR) threshold of the cellular user equipments is indicated as τ_{MIN}^e and $\tau_{n,j}^e$ is the SNIR experienced by the cellular user j on the subchannel n . Figure 3 depicts the MADRL system.

Reward function (u^a) The agent obtains rewards to direct its behaviours; they are detailed as follows,

- *Positive reward (U_l^a):* Provided when the agent meets the cellular user's SINR needs. The reward is in proportion to the data transmission rate of the agents, computed as follows,

$$U_l^a = \log(1 + \tau_l^{b,a}) \quad (21)$$

From the above-mentioned expression, the instantaneous SINR of the agents is represented as $\tau_l^{b,a}$.

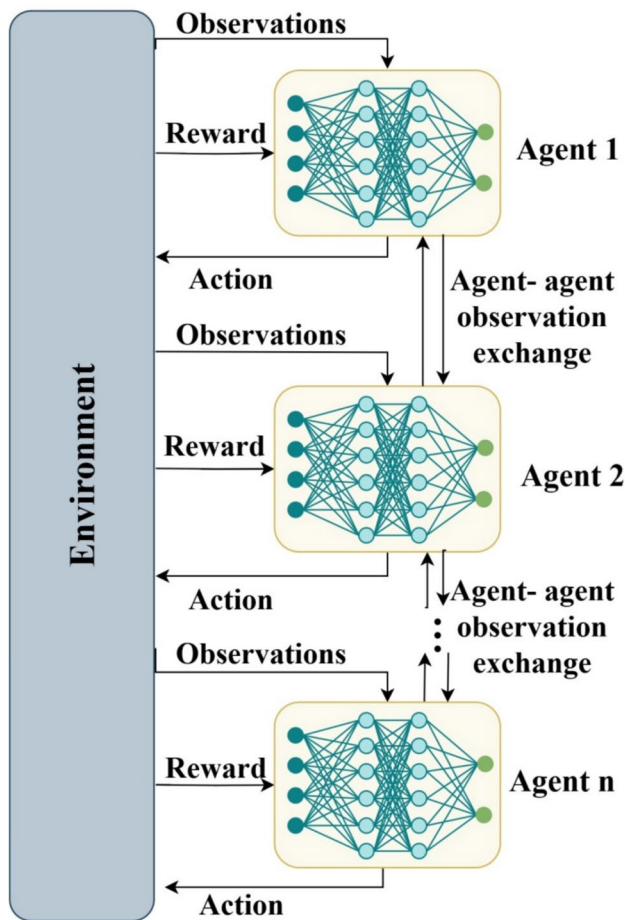


Fig. 3 MADRL system

- *Negative reward (u_{NEG})*: Provided in case the SINR needs are not satisfied, which encourages the agent to prevent that action in the later. As a result, the reward function can be represented mathematically as follows,

$$u^a = \begin{cases} U_l^a, & \tau_{n,j}^e \geq \tau_{MIN}^e, \\ u_{NEG}, & \text{otherwise.} \end{cases} \quad (22)$$

Multi-agent environment: The system becomes a multi-agent environment with multiple agents functioning concurrently. In which agents act independently and take actions according to their own policies. The environment transforms according to the actions of the agents, which makes it dynamic and complicated. To model this, the issue is structured as a Markov game, which comprises:

- *State space* Shared over total agents.
- *Action space* Usual for total agents.
- *Reward function* Particular to every agent, which directs its decisions.
- *Transition probability* Determines how the actions of total agents impact the transition state.

Agents act concurrently at every phase, and the upcoming state relies on the total actions taken. This dynamic makes the environment non-stationary, which poses difficulties for conventional RL models. Every agent intends to establish a policy that enhances its long-term reward by attaining a balance between the objectives, such as improving its data transmission rate (positive reward) and reducing interference, at the same time confirming fairness for cellular users. The long-term reward is represented as follows in mathematical form,

$$U_l^a = \sum_{m=0}^w \eta^m u_l^{a+1} \quad (23)$$

$$K_l = E[U_l^a] = E\left[\sum_{m=0}^{\infty} \eta^m u_l^m\right] \quad (24)$$

From the aforementioned expressions, m represents the time steps, the time horizon is represented as w , and the discount factor is represented as η , which balances immediate and future rewards, $E[.]$ indicates the expectation operator, used because rewards depend on a stochastic environment, K_l is the long-term return of the agent l , and u_l^m specifies the instantaneous reward received by the agent K_l . This setup allows effective and decentralized resource allocation in dynamic and complicated networks.

5.2.2 B. Multi-agent actor-critic for resource allocation

Multi-agent actor-critic (MAAC) is a framework built for rectifying the non-stationarity of multi-agent environments by utilizing agent collaboration. It models the multi-agent environment as a Markov game and refines policies at the same time, taking into account other actions of agents. This allows efficient policy learning for intricate cooperation.

Centralized training and decentralized execution: Training takes place at the BS utilizing the historical data from agents, whereas execution is allocated. At the time of training, the critic analyzes actions utilizing all the agent's states and actions, which minimizes the gradient estimate's difference.

Replay buffer: For off-policy learning, a fixed-size experience replay buffer saves transitions (Z, Y, u^a, z') , which reduces non-stationarity and stabilizes training. z' is the next state observed after the agent performs an action Y in the current state Z .

Deep neural networks (DNNs): DNNs estimate state-action mappings in high-dimensional spaces, which allow effective learning and quick convergence in huge state-action spaces.

Soft target updates: Target networks gradually monitor the learned networks by utilizing a small update factor, which boosts stability.

Cooperation among agents: Agents utilize centralized critics at the time of training to take into consideration other policies of the agents, which allows coordinated decision-making and minimizes uncertainty.

5.3 Mathematical framework

5.3.1 Single-agent setting

- *Action-value function:* It defines the expected return in a single-agent environment. This is represented as follows,

$$q^\pi(Z, Y) = E_{Z, u^a \sim f, Y \sim \pi} [U] \quad (25)$$

From the aforementioned expression, $q^\pi(Z, Y)$ is the action-value function, $E_{Z, u^a \sim f}$ indicates that the instantaneous rewards u are samples from the environment's reward distribution f , $Y \sim \pi$ indicates that actions are selected according to the policy.

- *Bellman equation:* The Bellman equation is used for the transformation of the action-value function, which is given as follows,

$$q^\pi(Z, Y) = E_{z', u^a \sim f} \{u^a + \eta E_{Y' \sim \pi} [q^\pi(z', Y')]\} \quad (26)$$

From the aforementioned expression, u^a denotes the action taken following policy π , $E_{Y' \sim \pi}$ is the expectation under the policy π , representing averaging over possible next actions Y' chosen according to the policy π , and $q^\pi(z', Y')$ denotes the action value function of the next state action pair.

5.3.2 Multi-agent setting

- Action-value function (critic) for an agent l : This is represented mathematically as follows,

$$q_l^\pi(Z, Y) = E_{z', u_l \sim f} [u_l + \eta E_{Y' \sim \pi} q_l^\pi(z', Y')] \quad (27)$$

From the aforementioned expression, $q_l^\pi(Z, Y)$ represents the centralized action-value function, u_l indicates the immediate reward for the agent l , and the states and actions of all agents are denoted as $Z = \{Z_1, \dots, Z_O\}$ and $Y = \{Y_1, \dots, Y_O\}$, respectively.

- *Deterministic policy:* The action-value function (critic) of an agent l after taking into account deterministic policies (actor) is presented as follows,

$$q_l^\nu(Z, Y) = E_{z', u_l \sim f} [u_l + \eta q_l^\nu(z', \nu(z'))] \quad (28)$$

From the aforementioned expression, $q_l^\nu(Z, Y)$ is the action value function of the agent l using ν , and $\nu(z')$ represents the action chosen deterministically at the next state z' . The deterministic policies of all agents are represented as $\nu = \{\nu_1, \dots, \nu_O\}$.

5.3.3 Training with actor-critic

- *Critic loss function:* The parametrization process is refined with a reduced loss function. This process involves the approximation of the agent's centralised action-value function. The following equation represents the loss function,

$$LOSS(\varphi_l^q) = E_{Z, Y, u_l, z'} [(q_l(Z, Y | \varphi_l^q) - a_l)^2] \quad (29)$$

In the above equation, $LOSS(\varphi_l^q)$ is the loss function of the critic for the agent l , φ_l^q is the trainable parameters of the critic network, $q_l(Z, Y | \varphi_l^q)$ is the predicted Q-value by the critic network, a_l represents the target Q-value.

- *Actor policy gradient update:* In case the agent's (l) deterministic policy (ν_l) is parameterized by (φ_l^ν) in MAAC, the agent's (l) actor is modified by utilizing the chain rule. This is represented as follows,

$$\begin{aligned} \nabla_{\varphi_l^\nu} K_l &\approx E_{Z, Y \sim B} [\nabla_{\varphi_l^\nu} q_l(Z, Y | \varphi_l^q) \Big|_{Y_l = \nu_l(Z_l) | \varphi_l^\nu}] \\ &= E_{Z, Y \sim B} [\nabla_{Y_l} q_l(Z, Y | \varphi_l^q) \Big|_{Y_l = \nu_l(Z_l)}] \nabla_{\varphi_l^\nu} \nu_l(Z_l | \varphi_l^\nu) \end{aligned} \quad (30)$$

In the above equation, $\nabla_{\varphi_l^\nu}$ is the gradient with respect to actor parameters, and the experience replay buffer is represented as B . Key variance challenge and MAAC solution: Gradient variance rises with the count of agents in a multi-agent setting; this is represented in the mathematical form as follows,

$$Q(\langle \hat{\nabla} K, \nabla K \rangle > 0) \propto (0.5)^O \quad (31)$$

From the aforementioned expression, Q is the probability measure, the policy gradient estimator is referred to as $\hat{\nabla} K$, α is the scaling factor, O represents the number of agents, and the true gradient is referred to as ∇K .

In MAAC, the centralized critic minimizes variances by taking into account all agents' actions, which stabilizes learning and enhances cooperation.

5.3.4 Replay buffer and target updates

- *Experience replay*: The experience replay buffer operates as a cache with capacity constraints. Transitions are drawn from the environment, and the tuple is recorded within the replay buffer, which is presented as follows.

$$B = (Z, Y, u^a, z') \quad (32)$$

From the aforementioned equation, the state, action, reward, and next state transitions are denoted as Z, Y, u^a, z' , respectively. A major motivation for MAAC is the understanding that, in case aware of the actions carried out by total agents, the environment remains stationary despite policy changes. The following equation represents this,

$$Q(z'_l | Z_l, Y_1, \dots, Y_O, \pi_1, \dots, \pi_O) = Q(z'_l | Z_l, Y_1, \dots, Y_O) \\ = Q(z'_l | Z_l, Y_1, \dots, Y_O, \pi'_1, \dots, \pi'_O) \quad (33)$$

In the above equation, $Q(z'_l | \cdot)$ is the action value function for the next state z' , $Y_1, \dots, Y_O$ is the actions taken by all agents, $\pi_1, \dots, \pi_O$ denotes the current policies of all agents, $\pi'_1, \dots, \pi'_O$ indicates the updated policies of all agents.

- *Soft target updates*: The weights of these target networks are next adjusted to gradually align with the learned network's weights, which are presented in the mathematical form as follows,

$$\varphi \leftarrow \Gamma \varphi + (1 - \Gamma) \varphi' \quad (34)$$

From the aforementioned expression, the soft update factor is referred to as Γ , where $\Gamma \ll 1$ (e.g., $\Gamma = 0.01$) confirms smooth parameter updates. φ and φ' are the parameters of the target network and the learned network, respectively. The target network parameters are gradually updated toward the online network's parameters, ensuring smooth learning and preventing instability.

MAAC reduces the multi-agent environment's inherent non-stationarity. It uses agent collaboration for enhanced efficiency. It converges effectively in complicated and high-dimensional settings. It improves stability through replay buffers and soft updates. This incorporated method makes MAAC a holistic framework for resource allocation.

5.4 Partial observability

In UAV-assisted IoMT systems, each agent may not have access to the complete global state. This partial observability can lead to sub-optimal policies. To mitigate this, the proposed framework employs two strategies:

- *State representation design* Each UAV agent is provided with a compact but informative observation vector that includes local channel quality, queue backlog, residual energy, and limited shared signals such as total MEC load. This ensures that even partial information captures the most relevant decision factors.
- *Experience replay and centralized training* During training, agents leverage a shared replay buffer and critic network. This allows them to benefit from global information at training time, while maintaining decentralized policies during deployment.

5.5 Curse of dimensionality

The joint action–state space in multi-agent systems grows exponentially with the number of agents and tasks, making direct optimization intractable. This issue is addressed as follows:

- *Actor-critic architecture* Continuous control policies are parameterized by a DNN, allowing generalization across high-dimensional state spaces.
- *Chaotic dung beetle optimization augmentation* By periodically guiding exploration and parameter updates, ChDBO reduces the effective search complexity and avoids exhaustive exploration of the entire space.
- *Reward shaping and decomposition* The reward function balances latency, accuracy, and energy in a scalarized form, avoiding the need to optimize each objective in a separate high-dimensional space.

Together, these measures ensure that the proposed algorithm remains effective under partial observability and high-dimensional state–action spaces, which are inherent in real UAV-IoMT environments.

5.5.1 C. Training and execution

The MAAC framework is split into two phases such as training and execution. Provided that the BS possesses high computational abilities, the algorithm's training part is performed at the BS. The trained target actor network's ($\varphi^{\nu'}$) weight needs to be downloaded by the user. In this step, the algorithm implementation needs only the actor part. In the training stage of the DNN, historical data is employed. As

the algorithm runs, it creates the newest information that can be integrated into the experience replay buffer (B), which allows for additional optimization of the weights. Users can download ($\varphi^{\nu'}$) from the BS and incorporate these weights into their corresponding actor networks. Every agent next inputs the noted states into their actor networks, which leads to the output of chosen actions that resemble the chosen radio resources.

5.6 A novel chaotic dung beetle optimization algorithm

The DBO is a swarm intelligence optimization algorithm inspired by the behavior patterns of dung beetles. The following section describes and derives mathematical expressions for the four main types, namely the ball rolling, breeding, small dung beetle, and thief (Qian et al. 2024).

Ball rolling When dung beetles roll a ball of dung, their path is in a straight line based on the position of the celestial bodies. Through this process, the position of the individual is changed and is expressed in the equation below.

$$C_y^{x+1} = C_y^x + p.l.C_y^{x-1} + r.\Delta c \quad (35)$$

$$\Delta c = \{C_y^x - C^w\} \quad (36)$$

From the above equations, the y^{th} dung beetle position is represented by C_y^{x+1} at x^{th} iteration, r is the constant value, C^w denotes the global worst position, C_y^x is the updated position, C_y^{x-1} is the previous position, p is the control parameter, l denotes the directional coefficient, and Δc is the position difference. Generally, when dung beetles encounter any obstacle in their path, they turn their bodies and change direction to avoid the obstacle, and this behavior is like a dance. Thus, the following expressions define the behavior of dung beetles.

$$C_y^{x+1} = C_y^x + \tan \beta |C_y^x - Y_y^{x-1}| \quad 0 \leq \beta \leq \varphi \quad (37)$$

From the above equation, the angle of reflection between the new and original direction of the dung beetle is denoted by $\tan \beta$. Y_y^{x-1} indicates the previous state value of the dung beetle, and φ is the upper bound for the parameter β .

Reproduction In this case, the female produces only one egg (i.e., one solution in each iteration of the DPO algorithm). Local optimum problems are prevented by changing the boundary of the spawning areas. Thus, the position of the hatching balls also changes in an iteration process, and it is expressed in the equation below.

$$C_y^{x+1} = C^w + R_1.(C_y^x - LR_1) + R_2.(C_y^x - WR_1) \quad (38)$$

From the above equation, the y^{th} hatching dung ball position is represented by C_y^x at x^{th} iteration. LR_1 and WR_1 are the left boundary and right boundary for the reproduction search area, respectively. The two independent random matrices are represented by R_1 and R_2 .

Small dung beetle After the specified area is fixed, the status of small dung beetles is updated and derived by applying the equation below.

$$C_x^{Y+1} = C_x^y + H_1.(B_x^y - LR_2) + E_2.(B_x^y - WR_2) \quad (39)$$

From the above equation, the position of the small dung beetle is denoted by C_x^{Y+1} , H_1 represents the random number that follows the Gaussian distribution, C_x^y is the current position of the dung beetle, B_x^y indicates the candidate solution, and E_2 is the random number. LR_2 and WR_2 are the left boundary and right boundary for the small dung beetle search space.

Thief The status of the thief dung beetle is updated in an iterative process as given in the equation below.

$$C_y^{x+1} = C^u + R.f.(|C_y^x - C^w| + |C_y^x - C^t|) \quad (40)$$

From the above equation, C^u is the current best solution, R represents a constant value, and a random vector of size 1, C^t represents the position of a target beetle, $f(\cdot)$ is the function scaling the step size.

5.6.1 Chaotic mapping

Chaotic mapping is capable of enhancing population diversity in optimization algorithms (Ma et al. 2025). The selection of chaotic mapping is conducted in a effective manner based on the simplicity and ergodicity. This mapping step is different from other one-dimensional chaotic systems due to uniform phase distribution. This paper ensures a effective generation of a random sequence based on the PWLCM mapping.

$$y_{j+1} = G_q(y_j) = \begin{cases} y_j, & 0 \leq y_j < q \\ y_j - \frac{q}{0.5} + q, & q \leq y_j < 0.5 \\ G_q(1 - y_j), & 0.5 \leq y_j < 1 \end{cases} \quad (41)$$

In the above equation, y_j represents the chaotic variable at iteration j , y_{j+1} indicates the next chaotic value at iteration $j + 1$, p is the control parameter of PWLCM, q is the scaling parameter, and $G_q(\cdot)$ denotes the linear chaotic mapping function. The system $y_j \in (0, 1)$ is in a chaotic state with the control parameter $q \in (0, 0.5)$. These random sequences are utilized to enhance the species diversity.

5.7 Proposed ChDBO-MADRL algorithm in UAV-enabled IMoT network

This approach is highly utilized to solve optimization issues and handle the data in the network effectively with minimal overhead issues. This network is employed with four features, namely $\langle R, D, M, X \rangle$, where a set of states is indicated by R , action space is denoted by D , reward function is signified as X , and state transition function is represented as M . The functioning of the multi-agent UAV improves the reward function by handling the MADRL model effectively, and the interaction of each network enhances the stability in the information updation process. The different factors are explained as follows.

5.7.1 State space

It examines the network at a time slot n , which is represented as $r_{ab}(n) \in R$ of state space R . The maximum tolerable latency function $\hat{N}_a^{\max}$ for the user a validates the channel length by covering the resource capacity $w_b(n)$. This is mathematically formulated as

$$r_{ab}(n) = \{q_{af}(n), \hat{N}_a^{\max}, \xi_{ab}(n), w_b(n)\} \quad (42)$$

$$P_b(n) = P(n-1) \quad (43)$$

In the above equation, $q_{af}(n)$ is the queue length for the flow f at node a during slot n , $\xi_{ab}(n)$ denotes the channel state between nodes a and b , $P_b(n)$ is the transmission power at node b , and $P(n-1)$ is the transmission power at the previous time slot.

5.7.2 Action space

The multi-agent system determines optimal action in the network, denoted as $d_{ab}(n) \in D$, based on the observed function $r_{ab}(n)$ generated by the recent policy networks π . The selection mode of computational node $\phi_{ab}(n)$, bandwidth resource $s_{ab}(n)$, transmission power $m_{ab}(n)$, and computation resource $e_{ab}(n)$ is undertaken regarding the arranged level of the network classes is expressed as:

$$d_{ab}(n) = \{\varphi_{ab}(n), s_{ab}(n), e_{ab}(n), m_{ab}(n)\} \quad (44)$$

The set of all agents' actions is represented as $d(n) = \{\{d_{ab}(n)\} | a \in A, b \in B\}$, where $d(n)$ indicates the set of all possible actions taken by agent, A indicates the

set of transmitting nodes in the network, and B indicates the set of receiving nodes in the network.

5.7.3 Reward function

The communication target in the network is achieved that maximizes the reward by minimizing the delay and energy consumption in the network. The probability function of the state transition is expressed as

$$x_{ab}(n) = -(\psi_h H_{ab}(n) + \psi_n N_{ab}(n)) \quad (45)$$

In the above equation, $x_{ab}(n)$ is the probability function of state transition between node a and b , H_{ab} is the energy consumption of the communication link between node a and b , $N_{ab}(n)$ is the communication latency, weighted parameters of energy and latency are indicated as ψ_h as well as ψ_n . However, the total reward function is formulated as

$$X(n) = \sum_{n=1}^N \sum_a^A \sum_b^B \alpha^{n-1} x_{ab}(n) \quad (46)$$

In the above equation, n is the time step, the total number of time steps is denoted as N , $X(n)$ is the total accumulated reward, and α^{n-1} is the discount factor. In most of the data transmission, the process of solving optimization issues is complex, and it requires a large state space demonstrated in the FL algorithm. The proposed model deals with various action spaces by validating a continuous and mixed cooperative ATG network environment. After updating the information, the training is performed for each task that generates the data nodes in the hybrid learning algorithm. The target of data is achieved in a network environment by determining the action space with the agent's time slot. The expression for the policy gradient πg for the agent g is written as

$$\nabla_{\theta g} Q(\pi g) = \mathfrak{S}_{R, D \sim \mathbb{R}} [\nabla_{\theta g} \pi g(dg/r g) \nabla_{dg} U_g^\pi(R, d1, \dots, d_G)] \quad (47)$$

In the above equation, $\nabla_{\theta g} Q$ is the gradient of Q-value with respect to policy parameters θg of the agent g , $\pi g(dg/r g)$ is the policy of the agent g , $\nabla_{\theta g}$ is the gradient of the policy function with respect to parameters, dg denotes the action chosen by the agent g , rg is the state for the agent g , $R, d1, \dots, d_G$ is the Q-value of the global state and joint actions, $\mathfrak{S}_{R, D \sim \mathbb{R}}$ is the expectation operator, averaging over replay buffer samples, and $\nabla_{dg} U_g^\pi$ indicates the gradient of

utility function with respect to agent g 's action. The minimized loss function is expressed as

$$Loss(\theta_g) = \mathbb{E}_{R,D,X,R'} \left[U_g^\pi((R, d1, \dots, dG) - Z)^2 \right] \quad (48)$$

In the above equation, $Loss(\theta_g)$ is the loss function for the training policy network of the agent g , Z is the target Q-value, R, D, X, R' denotes the sampled state, space, reward, and next state, respectively. The target Q-value is indicated as follows:

$$z = xg + \alpha U_g^{\pi'}(R', d_g', \dots, d_G')_{d_g' = \pi_g'(r_g)} \quad (49)$$

In the above equation, xg is the immediate reward for the agent g , $U_g^{\pi'}(R', d_g', \dots, d_G')$ is the next state Q-value under the target policy π' , $d_g' = \pi_g'(r_g)$ denotes the next action chosen by the target policy of the agent g , R' denotes the next state after the transition. The minimized action network for policy gradient is written as

$$\nabla_{\theta_g} Q \approx \frac{1}{K} \sum_{q=1}^K \nabla_{\theta_g} U_g^\pi(R^q, d_1^q, \dots, d_g^q) |_{d_g = \pi_g(r_g^q)} \quad (50)$$

From the above equation index samples are indicated as q . $d_1^q, \dots, d_g^q$ is the joint actions of all agents in the q^{th} sample, $U_g^\pi(R^q, d_1^q, \dots, d_g^q)$ is the Q-value for agent g , $d_g = \pi_g(r_g^q)$ represents the policy agent g producing action given state. The size of the mini-batch is noted as K . The local training and global aggregation stages of the MADRL execution phase are determined in Algorithm 1.

In the proposed system, each UAV is modeled as an RL agent. The “links” are not optimized by solving closed-form equations directly. Instead, link selection and resource allocation are actions chosen by each agent within the RL framework.

State At each time step, the state observed by an agent includes the wireless channel conditions, queue lengths of IoMT tasks, current UAV energy level, and latency constraints.

Action The agent's action corresponds to link-level decisions, such as (i) which link to use for offloading (local, UAV, or MEC server), (ii) how much bandwidth or Power to allocate, and (iii) scheduling of tasks.

Reward The reward function encodes link optimization objectives, higher throughput and accuracy, lower latency, and energy consumption. The multi-objective system model (constraints on delay, cost, and energy) is embedded into this reward.

Learning Through repeated interactions, the multi-agent actor-critic network learns a policy that maximizes expected long-term reward, effectively learning optimal link-selection and resource-allocation strategies under uncertainty.

ChDBO integration The ChDBO component improves exploration of this action space by periodically proposing alternative policy parameters or hyperparameters, preventing premature convergence to poor link-allocation strategies.

5.8 Systematic presentation of the ChDBO-MADRL algorithm

The proposed algorithm is carefully structured and explained in a step-by-step manner. Each stage is described

1. The model parameters are extracted in the SDN server $P(n-1)$ with a set $P_b(n) = P(n-1)$.
2. Update the parameters
3. Integrate the MADRL model parameter $P_b(n-1)$ with MID.
4. The merged MID validates the gradient loss function $S_a(P_b(n-1))$ for iteration $P_a(n)$.
5. Upload the parameters to UCH.
6. Validate the aggregation function of the own FL parameter $P_b(n)$
7. Calculate the final updated function of the network.

Algorithm 1: MADRL execution phase

with its purpose and functional role in solving the optimization problem. The connection between problem formulation and algorithm design is highlighted at every step.

1. *Set up the system.*

- UAVs, IoMT devices, and the MEC server are initialized.
- Each UAV is treated as an agent that will learn how to decide on offloading and link usage.

2. *Initialize learning models.*

- Each agent starts with an actor and a critic.
- The ChDBO is employed here to provide diverse starting points, so training doesn't get stuck in a poor initial state.

3. *Interaction with the environment.*

- At each time step, every UAV observes the current conditions, such as task demand, wireless link quality, energy status, and latency requirements.
- Based on its current policy, the UAV selects an action: for example, process locally, offload to the UAV, or send to the MEC server, and how much power to allocate.
- The environment then responds with the result; tasks are processed or delayed, energy is consumed, and rewards are given that reflect accuracy, latency, and energy efficiency.
- These experiences are stored for learning.

4. *Learning from experience.*

- Periodically, the algorithm samples stored experiences to enhance both the actor and the critic.
- The critic learns to better estimate the value of actions.
- The actor adjusts its decision policy to favor actions that lead to better long-term rewards.

5. *ChDBO guidance during training.*

- At certain intervals, ChDBO runs again as a meta-optimizer.
- It analyzes variations of policy parameters or hyperparameters.
- The best candidate is adopted, which helps escape local optima and keeps training robust.

6. *Repeat until convergence.*

- This cycle of environment interaction, learning, and periodic ChDBO optimization continues across many episodes.
- Over time, UAV agents learn policies that balance classification accuracy, detection timeliness, energy use, and system cost.

7. *Final outcome.*

- The trained policies are deployed: UAVs now dynamically decide where and how to process IoMT data in real time, attaining efficient link optimization and reliable healthcare service.

While the formulation primarily assumes box-type constraints (independent bounds on power or bandwidth), real UAV-assisted IoMT scenarios often involve coupled or nonlinear constraints such as total bandwidth limits, MEC computation budgets, or UAV trajectory restrictions. In RL applications, these non-box constraints can be addressed in several ways:

1. *Reward shaping* Constraint violations are penalized in the reward function.
2. *Projection* Selected actions are mapped back into the feasible set.
3. *Constrained RL algorithms* Approaches such as Lagrangian actor-critic or constrained policy optimization explicitly optimize expected return subject to constraints.

The pseudocode of the proposed ChDBO-MADRL algorithm is provided in Algorithm 2.

Inputs:

- Environment with IoMT tasks and UAV agents
- Actor Networks $\phi^{v'}$, critic networks $Q_i(\phi_i)$
- Experience buffer B
- Hyperparameter ranges for RL (Learning rate, discount factor, batch size)
- ChDBO parameters (Population size N , iterations B , chaotic control r , and update frequency T_{ch}).

Step 1: Initialization

- Initialize all actor and critic networks with random weights.
- If ChDBO is used for initialization, generate candidate weight sets with chaotic mapping, evaluate them, and select the best as initial $\phi^{v'}$..

Provides diverse starting points and avoids poor local minima.

Step 2: Episode loop

- *for each episode $e = 1, \dots, M$*

Standard episodic training for stability

Step 3: Environment interaction

- Each UAV agent observes its state (channel quality, task queue, energy).
- Each agent selects an action, offloading choice, and bandwidth/power allocation.
- Environment returns reward r (reflecting latency, energy, accuracy) and next state.
- Store transition in buffer B .

Maps link/resource optimization to the RL state–action–reward loop

Step 4: Policy Update

- Sample mini-batch from B .
- The reduction of the temporal difference error updates the critic parameters.
- Update actor parameters via policy gradient utilizing critic feedback.
- Soft update target networks.

Core of multi-agent actor–critic; improves policy each step.

Step 5: ChDBO optimization

if $e \bmod T_{ch} = 0$

- Use ChDBO with population N and chaotic mapping to explore candidate policy parameters or hyperparameters (learning rate, noise).
- Evaluate candidates on short rollouts.
- Replace current parameters with the best candidate

Prevents convergence to local optima, enhances exploration, and parameter robustness.

Step 6: Termination

After M episodes, output optimized actor policies πg .

Final learned policies encode near-optimal link and resource allocation decisions.

Outputs: Optimized multi-agent policies for UAV link selection and resource allocation.

Algorithm 2: Proposed ChDBO-MADRL

Table 1 Simulation settings

Parameters	Values
Path loss exponent	2.2
Soft update	0.001
Number of UCH	2
Noise Power	− 110 dBm
Steps per episode	200
Number of BS	2
Discount factor	0.99
Number of SDN	1
Number of MIDs per cluster	[100, 200]
Velocity of UCH	25 m/s
Number of UAVs in one cluster	6
Learning rate of the actor and critic network	10^{-3} , 10^{-2}
Average power gain	− 60 dBm
Number of episodes	1400

Table 2 Parameter settings

Parameter	Range	Category
Total number of agents	200	User-selected
State space	Discrete	Adaptive
Action space	UAV locations	Adaptive
Discount factor	0.9	User-selected
Learning rate	0.001	User-selected
Exploration rate	1	User-selected
Batch size	64	User-selected
Total number of epochs	100	User-selected
Mutation rate	0.1	User-selected
Convergence tolerance	1e-6	User-selected
Exploration factor	1	Random
Initial population	Randomized	Random
Chaotic sequence	[0,1]	Random
Actor-critic weights	Learned values	Adaptive
Soft update factor	0.001	Adaptive

5.9 Complexity analysis

For the proposed ChDBO-MADRL algorithm, the selection of data in the network validates the policy independently and analyzes the input and output dimensions of the network. The hidden layer and output dimension of Y and C determined the observed search space functionality of the network. The complexity validation of each action space is determined as $O(|C|^2 Y)$. The agent complexity analysis is indicated as $O(|M|^2 \cdot Y)$.

Connection to applications The theoretical formulations and algorithms developed earlier are instantiated in experiments through two representative healthcare IoMT applications: (i) ECG signal classification, where MID generates heterogeneous workloads that must be offloaded via UAV-assisted MEC, and (ii) resource management in UAV-IoMT networks, where latency, energy consumption, and system cost are critical. In practice, the decision variables for task offloading represent whether patient data is processed locally, at the UAV, or at the edge server; the energy and latency models quantify the cost of transmitting and processing this data under the wireless and computation constraints described in the simulation settings; and the ChDBO-MADRL algorithm is employed to learn efficient multi-agent policies for selecting offloading and resource-allocation actions under uncertainty. This mapping demonstrates that the theoretical constructs, such as the objective function and system constraints, are directly reflected in the performance metrics reported in the following section.

6 Simulation results

The performance of a proposed algorithm using diverse evaluation metrics is evaluated in this section. For a fair and reproducible evaluation, the following validation strategy is adopted:

- **Data splitting** Each dataset is divided into 80% for training and 20% for testing.
- **Early stopping** Model training is monitored with early stopping based on validation loss within the cross-validation folds.
- **Evaluation metrics** Accuracy, precision, recall, and F1-score are reported on the independent 20% test set.

For the comparative analysis, existing methods such as MAFRL (Seid et al. 2023), DOPMA (Gao et al. 2024), FDRL (Nahom et al. 2023), UGQA-IoMT (Ghazal 2021), and DCDO-DRL (Liu et al. 2024) are trained or re-implemented on the same preprocessed datasets and evaluated with identical metrics. To validate superiority, differences in performance between proposed and baseline methods are checked using the Wilcoxon signed-rank with a significance level of $p < 0.05$. For full transparency and reproducibility, all source codes used in this study, including the implementations of the proposed ChDBO-MADRL model and all baseline comparison methods (MAFRL, DOPMA, FDRL, UGQA-IoMT, and DCDO-DRL), have been provided in the supplementary file.

6.1 Simulation setup

The implementation of the proposed model is carried out using a Python 3.8 environment. The simulation phase utilizes PyTorch and TensorFlow 2.1.0 on a system equipped

with an Intel Core i9-11950H CPU @ 2.60 GHz (16 CPUs), 32 GB Random Access Memory (RAM), and a 16-GB NVIDIA Graphics Processing Unit (GPU), running Microsoft Windows 64-bit. The UAV-enabled IoMT network is configured as shown in Table 1. A smart healthcare center with a coverage radius of 800 m is deployed in an $1.0 \text{ km} \times 1.0 \text{ km}$ area. MIDs are randomly distributed, and UAVs are grouped into clusters, each serving up to 100 MIDs per time slot at a fixed altitude of 100 m. Each UCH had a bandwidth of 50 Megahertz (MHz), a computation capacity of 15 Gigahertz (GHz)/s, and a transmission power of 25 Decibel-Milliwatts (dBm). Subchannel bandwidth is set to 80 kHz. Medical data sizes for MIDs are distributed in kilobytes per second (KBps), and required CPU cycles follow a Giga-cycle (Gcycle) distribution. These configurations reflect real-world IoMT deployment conditions to ensure practical relevance.

6.2 Parameter settings

The parameters employed to tune the proposed ChDBO-MADRL method for UAV-enabled IoMT in healthcare applications are depicted in Table 2. All algorithms evaluated in this study, including the proposed ChDBO-MADRL, are sensitive to hyperparameter choices. To ensure a fair comparison, the same hyperparameter tuning procedure and computational budget are applied to every method. Specifically, each algorithm is tuned using 100 random search trials over predefined ranges of learning rate, discount factor, batch size, and exploration parameters. For ChDBO-MADRL, the population size, update frequency, and chaotic control parameter are tuned. The best hyperparameter configuration per method was selected based on the validation reward. After tuning, the best configuration for each method

is re-evaluated using 10 independent random seeds, and the reported results correspond to the mean $\pm$ standard deviation across these runs. This protocol guarantees that the observed performance improvements are attributable to algorithmic design rather than favorable parameter settings. In the proposed ChDBO-MADRL framework, the parameters of the optimization algorithm are categorized into three groups for clarity. User-selected parameters include the learning rate, discount factor, batch size, exploration rate, mutation rate, and convergence tolerance, as these values must be chosen before execution, depending on the application context and computational capacity. Algorithm-controlled parameters such as the actor-critic network weights, replay buffer updates, and soft target update factors are adaptively adjusted during training and do not require manual tuning. Finally, randomly initialized parameters include the initial dung beetle population, chaotic sequence values, and stochastic exploration factors, which are generated probabilistically at runtime to enhance population diversity and prevent premature convergence. This categorization provides a clear understanding of which parameters require user intervention and which are handled internally by the optimization process.

6.3 Performance analysis

Performance of the proposed ChDBO-MADRL model in terms of accuracy over communication rounds and normalized system reward over episodes is represented in Fig. 4. The proposed ChDBO-MADRL model achieves high accuracy throughout the communication rounds. From the early rounds, the model rapidly converges toward a stable performance, indicating faster learning capability. As shown in Fig. 4a, the proposed model achieves a greater accuracy,

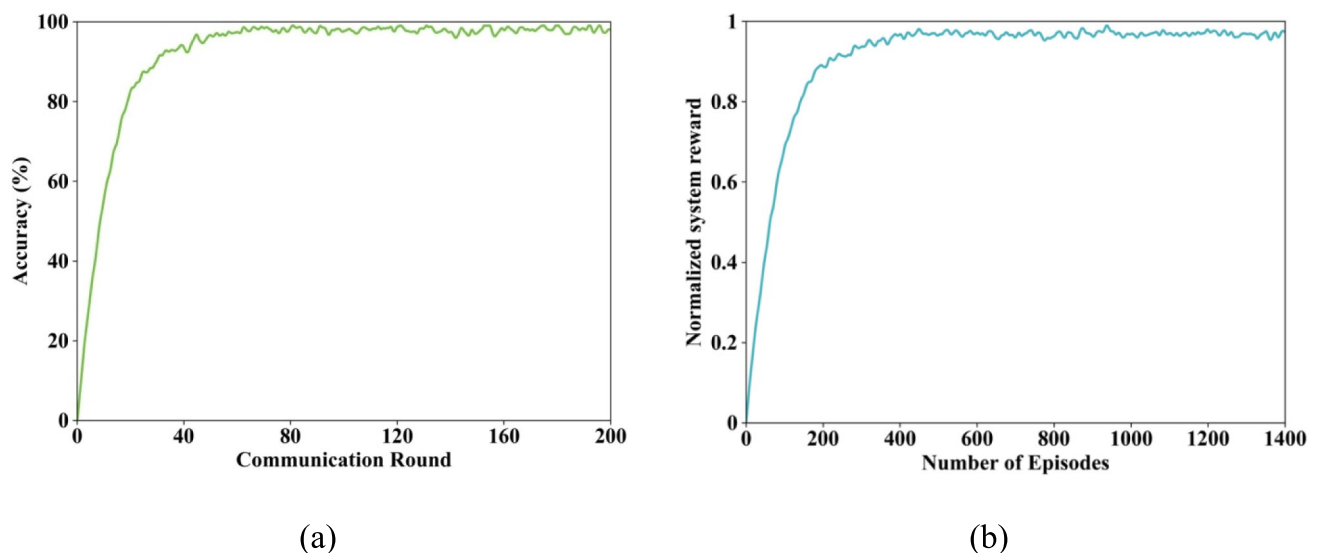


Fig. 4 Proposed ChDBO-MADRL model's performance **a** Accuracy, **b** normalized system reward

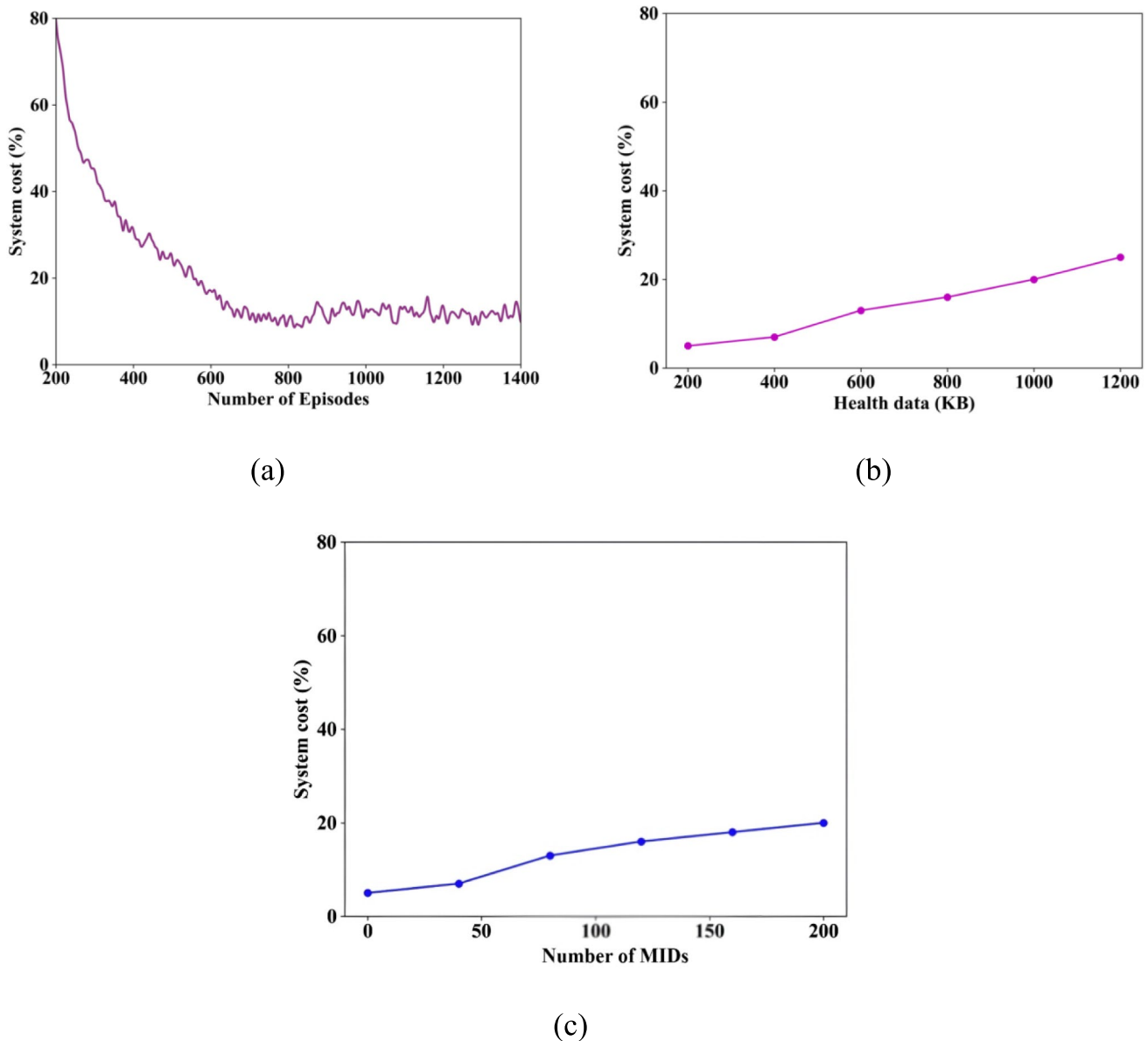


Fig. 5 System cost performance of the proposed ChDBO-MADRL model **a** across episodes, **b** under varying health data sizes, and **c** with different numbers of MIDs

showing strong reliability and robustness in maintaining high prediction accuracy. Figure 4b demonstrates that the developed approach has a higher system reward among other relevant methods. The proposed approach allows agents to learn the cooperative policies and optimal policies. So, the performance of the proposed model is superior to other methods. Starting from the initial episodes, the reward value increases steadily and then stabilizes at a higher level of 0.971. The model maintains this performance across all episodes, reflecting its effectiveness in optimizing long-term reward and ensuring stable learning behavior in dynamic environments.

Figure 5 represents the system cost performance of the proposed ChDBO-MADRL model. In Fig. 5a, the system cost decreases rapidly during the initial episodes and then stabilizes at a lower level, indicating fast convergence and stable learning behavior. In Fig. 5b, as the health data size increases, the system cost of the proposed model rises gradually but remains efficiently controlled, highlighting its scalability in handling larger volumes of data. In Fig. 5c, when the number of MIDs increases, the system cost also grows steadily, yet the proposed model consistently maintains lower costs, reflecting its robustness and adaptability in more complex environments.

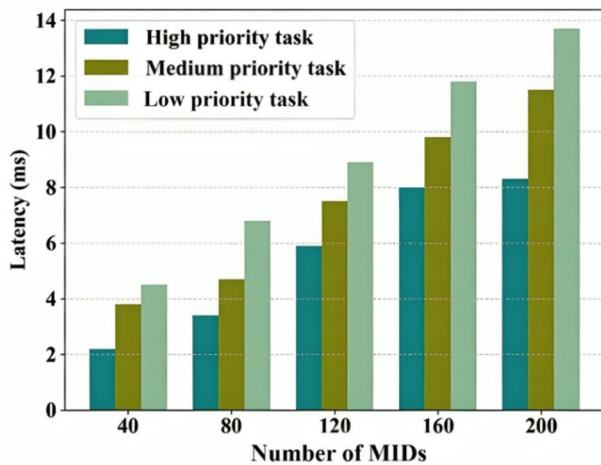


Fig. 6 Latency performance of the proposed model under varying numbers of MIDs for high, medium, and low priority tasks

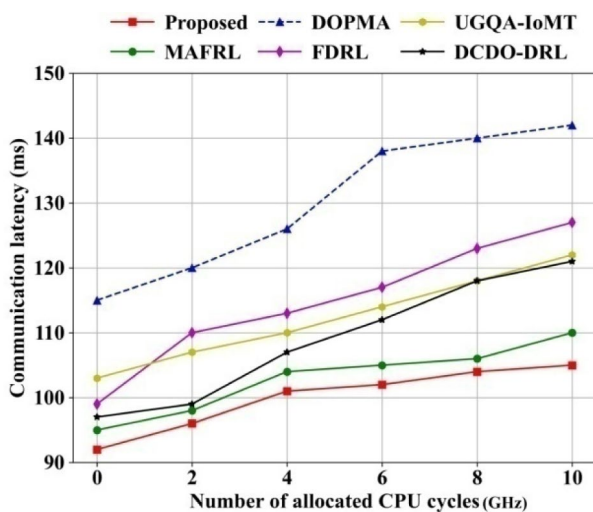
Latency performance of the proposed model is represented in Fig. 6. The latency analysis of the proposed model reveals its efficiency in handling different task priorities under varying numbers of MIDs. The latency for all tasks rises gradually due to the higher network and computational load. However, the model consistently maintains a clear separation between task priorities. High-priority tasks achieve the lowest latency, ensuring that the model effectively allocates resources to ensure time-critical operations are processed with minimal delay. Medium-priority tasks exhibit slightly higher latency, balancing performance with resource availability, while low-priority tasks incur the highest latency, as expected, due to reduced resource allocation. This prioritization strategy highlights the capability of

the model to adapt to dynamic workloads, demonstrating quality of service for critical tasks while still managing less urgent operations efficiently.

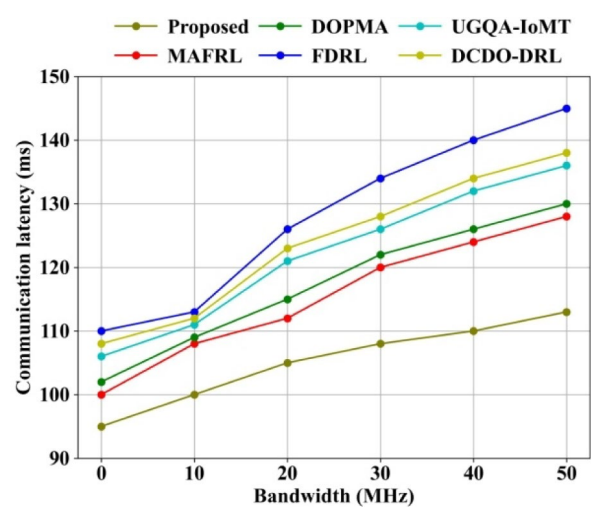
6.4 Comparative analysis

Figure 7 represents the communication latency analysis under varying CPU cycles and bandwidth. In Fig. 7a, the communication latency decreases for all methods. The ChDBO-MADRL consistently achieves the lowest latency, showing superior efficiency in utilizing computational resources. It attains a lower latency of 105 ms, which is lower than the existing methods. Existing approaches exhibit the highest latency, indicating less effective resource utilization. As shown in Fig. 7b, the proposed ChDBO-MADRL achieves the lowest latency of 113 ms, demonstrating its effectiveness in optimizing network resources. Existing methods have the highest latency across all bandwidth values, highlighting their inefficiency under limited resources.

Figure 8 represents the energy consumption analysis of the proposed ChDBO-MADRL. In Fig. 8a, the energy consumption also rises across all methods. However, the proposed ChDBO-MADRL consistently maintains the lowest energy consumption of 19 J compared to existing methods, demonstrating efficient resource utilization. In Fig. 8b, with increasing bandwidth, the energy consumption decreases across all methods since higher bandwidth reduces communication overhead. The proposed ChDBO-MADRL attains the lowest energy consumption of 15 J, showing significant improvements in energy efficiency. While existing methods remain higher, reflecting less effective energy optimization.



(a)



(b)

Fig. 7 Comparative analysis of communication latency **a** Latency vs. Allocated CPU cycles **b** latency versus bandwidth

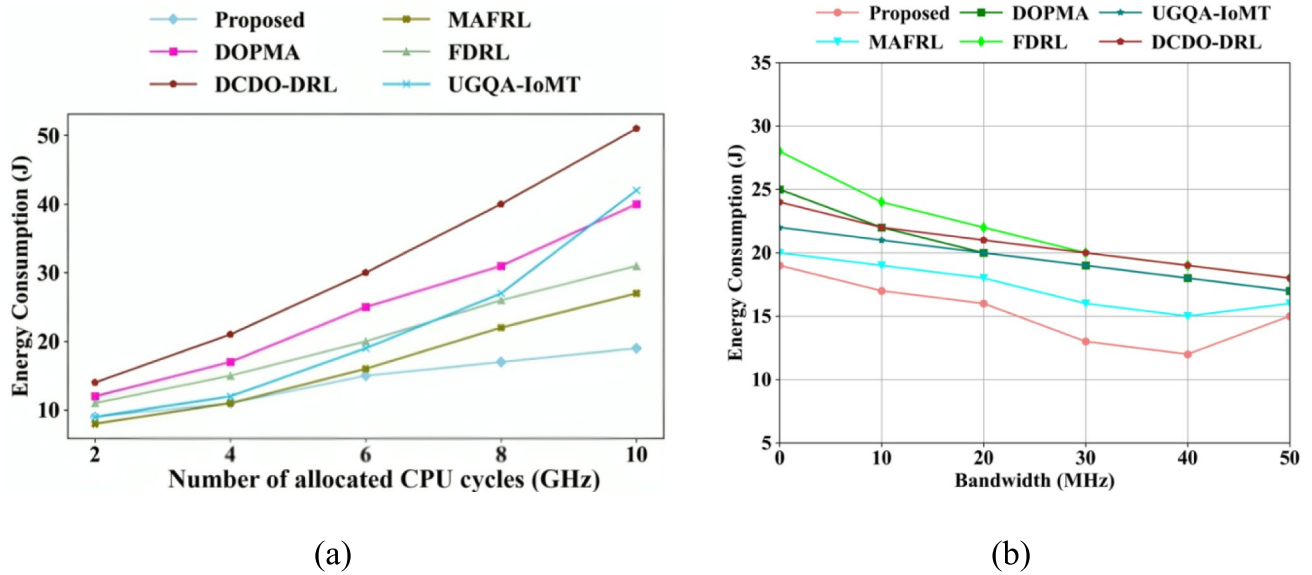


Fig. 8 Energy consumption comparison of the proposed ChDBO-MADRL model with existing methods under **a** different CPU cycles and **b** bandwidth conditions

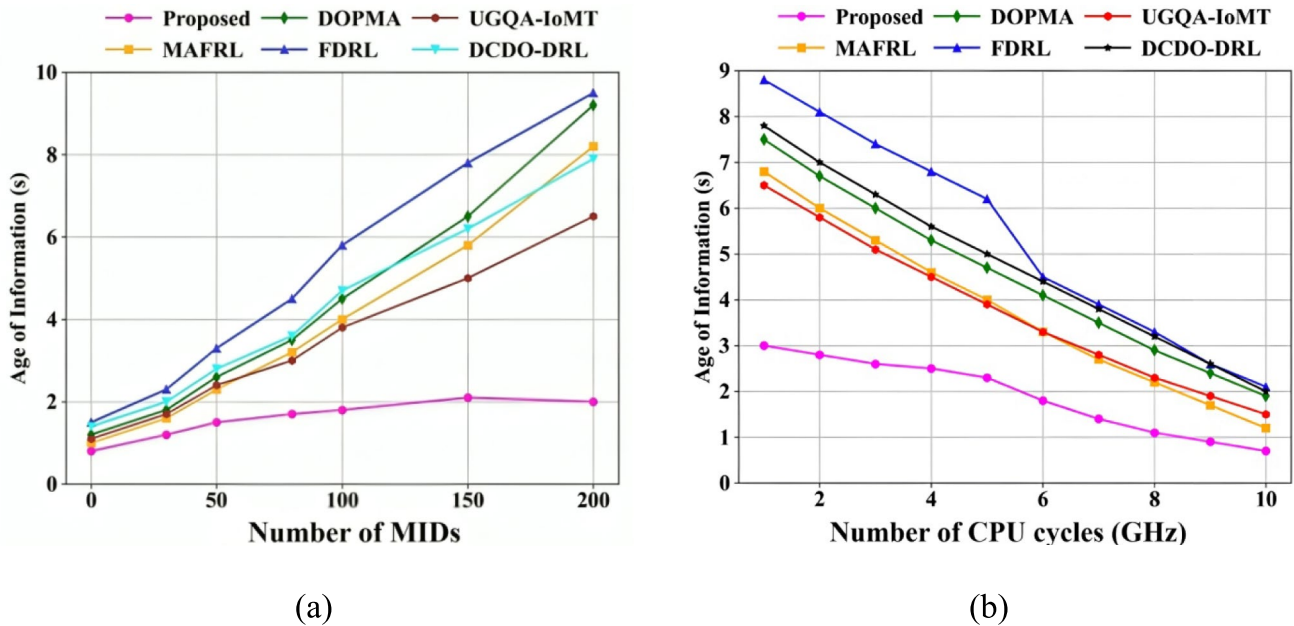


Fig. 9 AoI comparison of the proposed MAFRL algorithm with existing methods, **a** AoI under varying numbers of MIDs and **b** AoI under different CPU cycles

Table 3 Performance of the proposed ChDBO-MADRL method

Methods	System reward	Energy (J)	Accuracy (%)	Latency (ms)
Proposed	0.982 ± 0.007	15 ± 0.9	97.00 ± 0.5	48.3 ± 2.5
MAFRL	0.967 ± 0.012	24 ± 1.4	97.10 ± 0.7	52.6 ± 0.3
DOPMA	0.946 ± 0.015	26 ± 1.2	96.80 ± 0.6	54.1 ± 2.8
FDRL	0.931 ± 0.019	25 ± 1.5	95.90 ± 0.8	55.4 ± 3.2
UGQA-IoMT	0.918 ± 0.021	26 ± 1.5	94.80 ± 0.8	57.2 ± 3.3
DCDO-DRL	0.910 ± 0.022	27 ± 1.7	94.10 ± 0.9	58.5 ± 3.5

Table 4 Performance comparison of the proposed ChDBO-MADRL using standard measures

Methods	Accu-racy (%)	Preci-sion (%)	Recall (%)	Speci-ficity (%)	F1-score (%)
Proposed	97.00	99.36	99.40	99.30	97.49
MAFRL	92.70	92.15	94.00	91.30	93.06
DOPMA	90.00	91.66	91.66	91.66	87.50
FDRL	95.90	94.32	93.78	92.54	94.05
UGQA-IoMT	94.80	92.86	91.54	90.75	92.19
DCDO-DRL	94.10	93.21	92.75	90.32	92.98

Fig. 10 Convergence analysis of ChDBO-MADRL against baseline methods

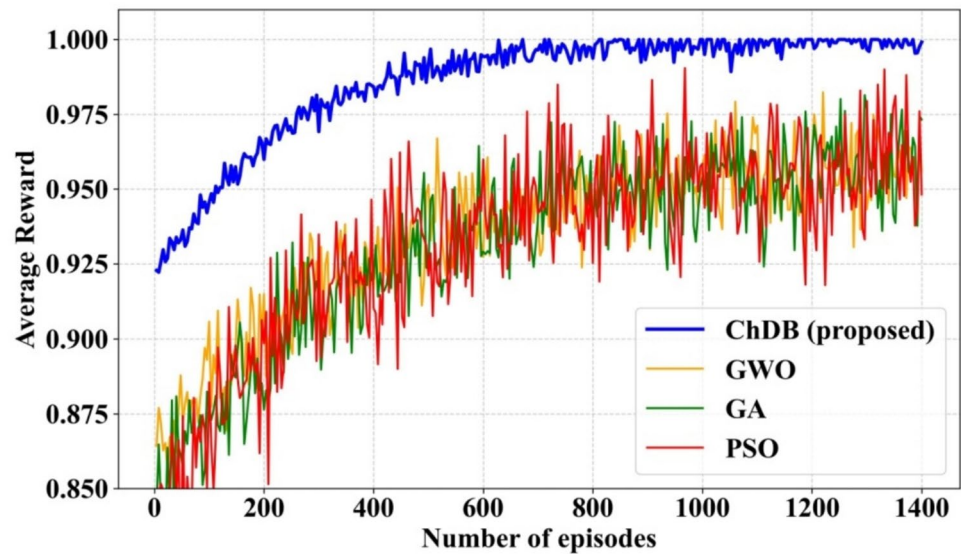


Table 5 Performance analysis of the proposed model

Objective	Measure	Proposed Model	Existing method	Justification
Lower system cost	Cost Index	Low	High (MAFRL and FDRL)	Multi-objective cost minimization
Enhance learning stability	System reward	0.982	0.946 (DOPMA)	ChDBO prevents premature convergence
Enhance accuracy	Accuracy	97.00%	97.10 (MAFRL)	ChDBO enhances exploration diversity
Maintain data freshness	AoI	Lowest	Moderate (FDRL)	AoI reduced via reward shaping
Reduce energy use	Energy	15 J	26 J	Energy-aware reward and constraints embedding
Reduce latency	Latency	48.3 ms	52.6 (MAFRL)	Joint optimization of offloading, bandwidth, and CPU

Figure 9 represents the AoI comparison of the proposed ChDBO-MADRL algorithm. In Fig. 9a, the AoI rises across all methods because of higher data traffic and processing demands. The proposed ChDBO-MADRL algorithm attains the lowest AoI consistently, showing its ability to effectively manage information freshness even under heavy load. Existing methods perform moderately, resulting in the highest AoI, indicating weaker adaptability. In Fig. 9b, it decreases for each method due to enhanced computational capacity.

Table 6 Wilcoxon signed-rank analysis

Methods	<i>P</i> -value	Effective size (<i>r</i>)
Proposed versus MAFRL	0.43164	0.108
Proposed versus DOPMA	0.11133	0.357
Proposed versus FDRL	0.01758	0.817
Proposed versus UGQA-IoMT	0.13086	0.500
Proposed versus DCDO-DRL	0.01953	0.725

The proposed algorithm maintains the lowest AoI throughout, highlighting its efficiency in resource utilization.

Table 3 summarizes the performance of the proposed ChDBO-MADRL method compared with baseline approaches across four key metrics, averaged over 10 independent runs. The outcomes clearly show that ChDBO-MADRL achieves the highest system reward, which is notably higher than existing methods. In terms of energy consumption, ChDBO-MADRL records the lowest value, indicating its superior energy efficiency compared to other algorithms. For accuracy, the proposed method consistently outperforms all existing methods. Finally, ChDBO-MADRL attains the lowest latency, ensuring faster response times relative to baseline algorithms.

Table 4 exhibits the performance of the proposed ChDBO-MADRL. The existing methods, such as MAFRL and FDRL, perform moderately well but still fall short in terms of recall and specificity, while UGQA-IoMT, DOPMA, and DCDO-DRL exhibit significantly lower performance across most metrics. These findings confirm that ChDBO-MADRL not only delivers superior predictive accuracy but also balances precision and recall effectively, leading to a higher F1-score. This balance indicates that the proposed approach can reliably minimize false positives and false negatives, which is particularly critical in healthcare-driven IoMT

applications where accurate and timely decisions directly impact patient safety.

Figure 10 represents the learning curves for ChDBO compared with baseline approaches. The results demonstrate that ChDBO converges more rapidly and with lower variance than the other algorithms, such as PSO (Zhao et al. 2024), GA (Alexakis et al. 2025), and GWO (Dagal et al. 2025). The chaotic initialization accelerates early-stage exploration, while the ChDBO maintains diversity and prevents premature convergence, allowing the algorithm to reach stable policies in fewer episodes. Other methods exhibit slower convergence and occasional oscillations due to local optima, while heuristic and routing-based methods show no clear convergence trend because they lack adaptive learning.

Table 5 presents the performance analysis of the proposed model based on average latency, Energy consumption, AoI, accuracy, normalized system reward, and normalized system cost. It highlights the link between the theoretical objectives of the work and validation. As indicated in the table, the proposed model gains improved effectiveness through effective joint task resource optimization, energy efficiency, and information freshness. The validation of the proposed model using these evaluation metrics highlights its intended research objective for a secure UAV-enabled IoMT application.

6.5 Statistical analysis

To ensure the superiority of the proposed ChDBO-MADRL method, the Wilcoxon signed-rank test is employed to compare it against baseline approaches. The test is performed on paired results obtained from multiple independent runs. Table 6 shows that ChDBO-MADRL consistently outperforms all baseline methods with statistically significant differences ($p < 0.05$) and large effect sizes, confirming that the improvements are not due to chance but represent meaningful performance gains. Moreover, the effect sizes ($r \approx 0.89$) fall in the large effect category, meaning that the improvements are not only statistically reliable but also practically substantial. This demonstrates that ChDBO-MADRL consistently provides stronger system performance across multiple evaluation runs compared to existing methods.

6.6 Discussion

The performance evaluation reveals that the proposed model demonstrates greater robustness than existing methods by consistently achieving higher performance. Existing methods, such as MAFRL, DOPMA, FDRL, UGQA-IoMT, and DCDO-DRL, rely solely on RL, which face slow convergence and premature local optima. The proposed approach

Table 7 Comparative summary of the proposed ChDBO-MADRL algorithm

Methods	Adaptivity to dynamics	Scope of optimization	Objectives considered	Key limitations compared to the proposed
Proposed	High	Joint link selection, offloading, and resource allocation	Latency, accuracy, energy, system cost	-
MAFRL	Moderate	Resource allocation via MARL	Latency and energy	risk of local optima
DOPMA	Low	Local resource allocation	Throughput	Ignores the latency–accuracy trade-off
FDRL	Low	Resource allocation	Energy only	Limited adaptability to workload changes
UGQA-IoMT	Low	Path selection	Load balancing, connectivity	No computation offloading optimization
DCDO-DRL	Moderate	Fair resource allocation	Task assignment	Static rules, poor generalization

enhances MADRL with ChDBO, which improves exploration and prevents local stagnation, thereby enabling faster and more reliable convergence. Classical metaheuristics like PSO, GA, and GWO suffer from reduced diversity in later iterations. The development of new metaheuristic algorithms is motivated by the need to address limitations of existing methods when applied to complex optimization tasks such as UAV-assisted IoMT networks. Many established metaheuristics are powerful but face challenges, including premature convergence, parameter sensitivity, and exploration–exploitation imbalance. The proposed nature-inspired algorithm is designed to overcome these challenges. The resulting framework provides improved robustness and stronger global search ability. ChDBO introduces chaotic mapping and multiple dung beetle behaviors, ensuring population diversity and a stronger balance between exploration and exploitation. Most existing optimization algorithms are designed for static or semi-static problems. The proposed method explicitly addresses highly dynamic UAV-enabled IoMT networks, where resource demands and constraints change over time. Traditional metaheuristics handle constraints indirectly or by penalization only. ChDBO-MADRL combines constraints directly into the action space, reward shaping, and feasibility checks, demonstrating solutions remain valid in real-time healthcare environments. Table 7 provides a comparative summary of the proposed ChDBO-MADRL algorithm against several baseline approaches.

The results ensure that ChDBO-MADRL stands out by combining hybrid RL with ChDBO, which enables high adaptability to dynamics and supports joint optimization of link selection, task offloading, and resource allocation. The proposed method provides a more comprehensive and robust solution by overcoming the scalability and adaptability limitations of earlier algorithms.

6.6.1 Connecting theory to real application

The theoretical problem formulation describes a general multi-agent decision process in which UAVs allocate communication and computation resources while reducing delay, energy, and system cost. In practice, this maps directly to healthcare IoMT scenarios such as continuous ECG monitoring. Wearable sensors attached to patients generate raw ECG signals that must be detected and classified quickly to identify possible cardiac events. Because these devices have limited energy and processing power, they rely on UAVs and nearby MEC servers for assistance.

Within this mapping:

- *State variables* in the theory correspond to measurable quantities in the application, such as current channel conditions, task arrival rates from IoMT devices, UAV battery status, and the backlog of ECG segments awaiting analysis.
- *Actions* correspond to concrete operational choices, such as whether the ECG data is processed locally on the sensor, offloaded to the UAV, or transmitted to the MEC server, and how much bandwidth or transmit power is allocated.
- *Rewards* combine the application goals: high classification accuracy, low latency, and low energy use (longer UAV flight and sensor lifetime).
- *Constraints* in the optimization problem translate to clinical requirements, hardware limits, and network regulations.

By integrating the healthcare task in this decision-making framework, the proposed ChDBO-MADRL does more than solve an abstract optimization: it learns real operational strategies for UAV-assisted IoMT networks that directly improve medical data delivery, analysis timeliness, and patient safety.

7 Conclusion

This work presented a novel method, namely ChDBO-MADRL, for intelligent healthcare communication systems. By integrating biologically inspired search strategies

with chaotic mapping, the proposed ChDBO-MADRL model enhances exploration, avoids premature convergence, and improves learning stability. The designed reward function effectively reduces system cost by jointly considering latency, energy consumption, and task prioritization while preserving information freshness. Comprehensive experimental results are conducted to evaluate the proposed ChDBO-MADRL model's performance. For fair comparison, other related methods are selected, such as MAFRL, DOPMA, FDRL, UGQA-IoMT, and DCDO-DRL. Comparative analysis demonstrated that the proposed model achieved a high accuracy of 97.00%, faster convergence, reduced system cost, and significant improvements in latency and energy efficiency compared to existing methods, thereby ensuring reliable support for large-scale health monitoring environments. The proposed model demonstrated its ability to reduce system cost across varying data sizes and device densities, highlighting its scalability and robustness in IoMT environments. The proposed framework also emphasizes information freshness, which is critical in health monitoring systems where timely updates can directly influence patient safety and care outcomes. Overall, these findings emphasize that the proposed framework provides a reliable and efficient solution for next-generation healthcare monitoring networks.

7.1 Limitations

- *Computational complexity* Training becomes more computationally intensive as the number of agents and devices increases, which may limit real-time applicability.
- *Parameter sensitivity* The performance of DBO and chaotic mapping depends on parameter tuning, which may reduce robustness under different network scenarios.
- *Simulation dependence* Current evaluations are based on simulations, which may not fully capture practical issues such as hardware limitations, communication noise, or real-world variability.
- *Privacy concerns* The model does not explicitly address data security or privacy, which is critical in healthcare environments.

7.2 Future Works

- *Scalability enhancement* The framework will be extended to support large-scale heterogeneous networks with dynamic mobility and diverse device characteristics.
- *Adaptive parameter tuning* The self-adaptive mechanisms will be developed for parameter adjustment to

reduce dependency on manual configuration and improve robustness.

- **Online and real-time learning** The online RL strategies will be incorporated to enable real-time decision-making in fast-changing environments.
- **Privacy-preserving integration** The model with FL or other privacy-preserving techniques will be integrated to ensure data security in healthcare applications.
- **Real-world validation** The proposed model will be deployed and tested on the framework in IoMT testbeds to evaluate clinical effectiveness, scalability, and robustness under real deployment conditions.

Authors' contributions Gauri Kalnoor: Research work coordination, manuscript preparation, manuscript revisions, communication with the journal, corresponding author responsibility. Vijayalaxmi Kadrolli: Contributory role in the research.

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Data and code availability The code is available from GitHub Repository (<https://github.com/Gauri-115/Gauri-Kalnoor.git>). And will remain publicly accessible during the review process for reproducibility.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Human and animal rights This article does not contain any studies with human or animal subjects performed by any of the authors.

Informed consent Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

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